

Copula-Based Mean-Reverting Processes: Theory and Evidence from Kraken USDT/USD

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Abstract

We develop a copula-based framework for mean-reverting Markov processes and evaluate its empirical content using five-minute Kraken USDT/USD prices. The theory distinguishes local drift reversion, finite-horizon conditional mean reversion, and convergence of moments under an initial law. For absolutely continuous Markov copulas, we derive conditional-mean and generator criteria, establish observable densities for monotone transforms of OU and CIR diffusions, and construct a stationary-reset family that preserves the invariant marginal while multiplying conditional-mean persistence by a survival probability. The empirical analysis separates mathematical, predictive, and structural validity. Margin-constrained mixed-OU models improve five-minute validation and test log scores relative to OU. An unrestricted conditional QML fit scores still better, but its invariant standard deviation is 5.76 times the empirical value and its implied 50-basis-point tail probability is economically implausible. Peg-constrained and inference-functions-for-margins estimates repair this scale failure, yet their Gaussian margins still miss the extreme kurtosis of the data. A latent-OU measurement-noise benchmark attributes 4.3 percent of stationary variance to observation noise and does not eliminate the tail discrepancy. A continuous-path boundary bootstrap provides a strong pilot signal against OU, but it has not yet reached the pre-specified 1,999-replication threshold for formal inference. The evidence therefore supports the mixed Copula as a coherent dependence mechanism, while requiring thick-tailed or semiparametric margins before it can be interpreted as a complete stationary structural model.

Keywords: mean reversion; Markov copula; stablecoin; transition density; stationary reset; invariant margin

JEL classification: C22; C58; G12; G15

1 Introduction

Mean reversion is a central organizing principle in continuous-time finance. The Gaussian specification of Vasicek (1977) and the square-root diffusion of Cox et al. (1985) remain canonical because they combine an interpretable restoring force with tractable transition laws. Empirical financial dynamics, however, are often nonlinear and state dependent. Chan et al. (1992) compare alternative short-rate diffusions, Aït-Sahalia (1996) documents nonlinear drift and volatility, and Bibby and Sørensen (1995), Kessler (1997), and Aït-Sahalia (2002) develop inference for discretely observed diffusions. These findings motivate models that retain a precise restoring mechanism without forcing the marginal distribution and temporal dependence into one Gaussian specification.

Copulas provide that separation. Darsow et al. (1992) connect Copulas and Markov transition operators through the star product, while Chen and Fan (2006) and Chen et al. (2009) develop semiparametric estimation for stationary Copula-based Markov models. Monotone space-time transformations provide a complementary construction (Bibbona et al., 2016), and Fang et al. (2020) study Copula-based Markov processes directly. This literature establishes how to separate margins from serial dependence. It does not make stationarity, a declining dependence coefficient, and an economic restoring force interchangeable. We therefore formulate

mean reversion through conditional drift and conditional-mean restrictions and state separately what is needed for convergence under a chosen initial law.

Stablecoins provide a sharp application because the state of interest is the deviation from a peg. The survey of Ante et al. (2023) documents a broad and rapidly growing empirical literature. Lyons and Viswanath-Natraj (2023) connect Tether’s peg stability to the design and accessibility of arbitrage, while Duan and Urquhart (2023) find that correction speeds vary across stablecoins and over time. Nonlinear or time-varying stabilization also appears in Djogbenou et al. (2023) and Hui et al. (2025); Bergault et al. (2024) model pegged assets with nested OU factors. Run incentives and tail events add another dimension (Ma et al., 2023; Liu et al., 2023; Ahmed et al., 2025; Eichengreen et al., 2025). A useful model must therefore represent ordinary correction and exceptional transitions without confusing a better short-horizon density fit with a credible long-run distribution.

The paper develops three process families. Section 2 gives an absolutely continuous Markov–Copula foundation and exact conditional-mean and generator criteria. Section 3 constructs observable processes through strictly monotone transformations of OU and CIR diffusions. The inverse map and Jacobian make explicit when a formal PDE solution does not define an estimable model because monotonicity, support, or invertibility fails. Section 4 introduces the central stationary-reset construction. A multiplicative mixture of a stationary transition Copula and independence is a genuine Markov evolution, preserves its invariant margin, and multiplies conditional-mean persistence by a survival probability. Its nonhomogeneous Poisson representation places the result within the stochastic-resetting literature (Evans and Majumdar, 2011; Evans et al., 2020); the contribution here is not the reset mechanism itself, but its Markov–Copula formulation, margin preservation, observable density, and exact reversion identities.

Section 5 evaluates these restrictions on Kraken USDT/USD from 2021 through 2025. The design estimates every theoretical family with an observable finite-dimensional density or retains a failure diagnosis. For the central mixed-OU family, it distinguishes a peg-constrained estimator (MOU-Peg), a Gaussian inference-functions-for-margins estimator (MOU-IFM), and unrestricted conditional QML (MOU-UQML). Every candidate is judged on three layers: kernel validity, out-of-sample conditional score, and compatibility of its invariant margin with the data. A latent OU plus observation-noise model addresses the concern that high-frequency price discreteness can distort apparent reversion (Holý and Tomanová, 2018).

This separation changes the empirical conclusion. MOU-UQML attains the best five-minute validation and test log scores among the four central transition models, but implies an invariant scale almost six times the empirical scale and a grossly excessive depeg probability. MOU-IFM and MOU-Peg improve on OU without this scale failure, but their Gaussian margins cannot reproduce the sample’s extreme kurtosis. Thus the data support a second, reset-like dependence channel; they do not yet validate a complete Gaussian stationary structural law. This conclusion is deliberately narrower than an in-sample likelihood ranking and aligns the claims of the paper with the restrictions proved in Sections 2–4.

2 Copula-Based Mean-Reverting Markov Processes

2.1 Probability space, supports, and notation

Work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions. Conditional laws given a point are always versions of regular conditional probability kernels in the sense of Kallenberg (2021, Chapter 8). Such kernels are probabilistically unique only up to null sets in the conditioning variable. Equalities involving an unspecified version are therefore asserted for the relevant marginal law almost everywhere; an explicitly selected pointwise version, or continuity of both sides, is needed before an equality is asserted at every interior state.

For a distribution function F , its generalized quantile is

$$Q(u) = F^{-1}(u) := \inf\{x \in \mathbb{R} : F(x) \geq u\}, \quad 0 < u < 1. \quad (1)$$

Quantiles are not assumed finite or differentiable at $u = 0, 1$. The support of the law at time t is denoted by $E_t = \text{supp}(F_t)$. A statement that a CDF is strictly increasing means strictly increasing on the interior of its support, not on all of $\mathbb{R}$. The symbol m denotes the target mean, $\bar{r}$ a base diffusion's long-run level, and s a Gamma scale. These objects were all denoted by θ in the original manuscript and must not be conflated.

2.2 Markov–copula foundations

2.2.1 Absolutely continuous Markov copulas

Let $I = (0, 1)$ and let Leb be Lebesgue probability measure on I . Set $\Delta = \{(s, t) : 0 \leq s < t\}$. We use the standard copula and star-product terminology of Nelsen (2006) and Darsow et al. (1992), subject to the pointwise-kernel qualifications below.

Assumption 2.1 (Absolutely continuous Markov–copula density evolution). *There is a selected transition-density version, namely a jointly Borel-measurable function $c : \Delta \times I^2 \rightarrow [0, \infty)$; write $c_{s,t}(u, v) = c(s, t, u, v)$. For every $s < t$,*

$$\begin{aligned} \int_I c_{s,t}(u, v) \, dv &= 1, & \text{for every } u \in I, \\ \int_I c_{s,t}(u, v) \, du &= 1, & \text{for Leb-a.e. } v \in I, \end{aligned} \quad (2)$$

$$c_{s,t}(u, v) = \int_I c_{s,r}(u, w) c_{r,t}(w, v) \, dw, \quad \begin{aligned} &0 \leq s < r < t, \quad u \in I, \\ &\text{for Leb-a.e. } v \in I. \end{aligned} \quad (3)$$

For each fixed (s, r, t, u) in (3), the exceptional Lebesgue-null set in v may depend on (s, r, t, u) . For $B \in \mathcal{B}(I)$, define the selected kernel

$$K_{s,t}(u, B) := \int_B c_{s,t}(u, v) \, dv, \quad u \in I, \quad s < t, \quad (4)$$

and put $K_{t,t}(u, B) = \mathbf{1}_B(u)$. Thus “for every u ” below refers to this chosen pointwise kernel version; it is stronger than merely specifying a two-dimensional density up to Leb^2 -null sets. For $s < t$, define

$$C_{s,t}(x, y) = \int_0^x \int_0^y c_{s,t}(u, v) \, dv \, du, \quad (x, y) \in [0, 1]^2. \quad (5)$$

On the diagonal set $C_{t,t}(x, y) = \min(x, y)$ and use the identity transition operator; no Lebesgue density $c_{t,t}$ is asserted.

Proposition 2.2 (Density consistency and the copula star product). *Under Assumption 2.1, (5) is a copula with density $c_{s,t}$. The kernels (4) satisfy, for every $u \in I$ and every $B \in \mathcal{B}(I)$,*

$$K_{s,t}(u, B) = \int_I K_{r,t}(w, B) K_{s,r}(u, dw), \quad 0 \leq s < r < t. \quad (6)$$

Consequently, the operators

$$(P_{s,t}g)(u) = \int_I g(v) c_{s,t}(u, v) \, dv, \quad P_{t,t} = \text{I}, \quad (7)$$

form a pointwise Markov evolution preserving the uniform law. In particular, the pair-copulas obey the star-product consistency relation

$$C_{s,t} = C_{s,r} * C_{r,t}. \quad (8)$$

More explicitly, if C, D have selected densities c_C, c_D , respectively, the density-based definition used here is

$$(C * D)(x, y) = \int_I \left\{ \int_0^x c_C(a, w) da \right\} \left\{ \int_0^y c_D(w, b) db \right\} dw. \quad (9)$$

This equals the familiar formula $\int_I \partial_2 C(x, w) \partial_1 D(w, y) dw$ when the ordinary partial derivatives are interpreted as measurable almost-everywhere representatives in the integration variable w . For every $0 \leq t_0 < \dots < t_n$, the finite-dimensional density

$$f_{t_0, \dots, t_n}(u_0, \dots, u_n) = \prod_{j=1}^n c_{t_{j-1}, t_j}(u_{j-1}, u_j) \quad (10)$$

is projectively consistent. Hence these densities define a Markov process U with uniform marginals on the continuous time index. No sample-path continuity is implied by this construction alone.

Proof. Nonnegativity and the two identities in (2) give groundedness, two-increasingness, and uniform margins. For fixed (s, r, t, u) , integrate (3) over B and use Tonelli's theorem to obtain (6); hence $P_{s,r}P_{r,t} = P_{s,t}$ pointwise. Another application of Tonelli gives (9) and (8). The first marginalization in (10) uses the column normalization, the last uses the row normalization, and removal of an intermediate time uses (3). Thus the family is projectively consistent. Since I is a standard Borel space, the Kolmogorov extension theorem supplies the stated process (Kallenberg, 2021, Chapter 8). The full integral details concerning derivatives and null sets are given in Appendix A.1. $\square$

Remark 2.3 (Ordinary derivatives and modified partial Dini derivatives). For the selected density version, define the pointwise conditional CDF

$$H_{s,t}^c(u, y) := K_{s,t}(u, (0, y]) = \int_0^y c_{s,t}(u, v) dv, \quad (u, y) \in I \times [0, 1]. \quad (11)$$

For each fixed y , the fundamental theorem for Lebesgue integrals gives

$$H_{s,t}^c(u, y) = \partial_1 C_{s,t}(u, y) \quad \text{for Leb-a.e. } u. \quad (12)$$

The modified partial Dini derivative $\mathfrak{D}_1 C$ of Fang et al. (2020, Definition 2.1 and Theorems 2.1–2.2) selects a CDF-valued kernel at every source state, including points where the ordinary partial derivative is absent. It and H^c are versions of the same regular conditional law and therefore agree for Leb-a.e. source state, but need not agree on a source-null set. If $u \mapsto H^c(u, y)$ is continuous at a given u —for example under an appropriate continuous, dominated density version—then the difference quotient for $C(\cdot, y)$ converges there and all three quantities agree at that u . Thus existence of a Radon–Nikodym density alone justifies (12), not an unrestricted pointwise equality. This is precisely why the present density-only framework selects (4) explicitly rather than silently identifying a density version with an everywhere-defined derivative; see also the differentiation results in Folland (1999, Chapter 3).

Remark 2.4 (Why some statements say “every u ” and others “a.e. v ”). The source coordinate u indexes the selected probability kernel $K_{s,t}(u, \cdot)$, so its row normalization and kernel Chapman–Kolmogorov equation are required for every u . The target coordinate v is a Radon–Nikodym density variable, and densities and their identities are intrinsically determined only Leb-a.e. in v . Integrating a density identity that holds for a.e. v gives a measure identity for every Borel

set B . Conversely, equality of two absolutely continuous kernels for every B identifies their densities only a.e. in v . These are roles of the two coordinates, not properties of the letters used to name them.

Star-product consistency of copula CDFs by itself recovers the conditional composition only for a.e. source state; it does not choose a pointwise transition function. The distinction, and sufficient regularity conditions that upgrade it, are made explicit in Theorems 3.1–3.2 of Fang et al. (2020). Assumption 2.1 imposes the stronger selected-version property directly.

Remark 2.5 (Scope of the density restriction). This article deliberately restricts attention to absolutely continuous pair-copulas because every copula used in the empirical likelihood has a density. This is a simplifying scope assumption, not a claim that all Markov copulas are absolutely continuous. Consistency (3), rather than the validity of isolated bivariate copulas at selected horizons, is what supports a continuous-time Markov model.

2.2.2 Uniformization and exact conditional-expectation identities

Assumption 2.6 (Regular marginal representation). *For every t , F_t is continuous and strictly increasing on E_t° , Q_t is finite on I , and $\int_I |Q_t(u)| du < \infty$. The process U has the transition densities in Assumption 2.1, is adapted to $(\mathcal{F}_t)$, and is Markov with respect to this filtration. Moreover,*

$$X_t = Q_t(U_t) \quad \mathbb{P}\text{-a.s.} \quad (13)$$

Theorem 2.7 (Exact quantile–density criterion). *Under Assumption 2.6, for $s < t$,*

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = \int_I Q_t(v) c_{s,t}(U_s, v) dv \quad \mathbb{P}\text{-a.s.} \quad (14)$$

Consequently, the integrable process X is a martingale with respect to $(\mathcal{F}_t)$ if and only if

$$Q_s(u) = \int_I Q_t(v) c_{s,t}(u, v) dv \quad \text{for each fixed } s < t \text{ and for Lebesgue-a.e. } u. \quad (15)$$

The exceptional set in u may depend on the time pair (s, t) . If both sides of (15) have continuous versions in u , the equality extends from a.e. u to every $u \in I$.

Proof. The Markov property of U with respect to $(\mathcal{F}_t)$ and (13) give (14). Since $Q_s(U_s) = X_s$ and U_s is uniform, the martingale identity $\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$ is equivalent to (15) up to Lebesgue-null sets. $\square$

Proposition 2.8 (Observable transition density). *If, in addition, for every t , F_t is absolutely continuous with density $p_t > 0$ on E_t° , define, for every $x \in E_s^\circ$,*

$$K_{s,t}^X(x, B) := \int_I \mathbf{1}_B(Q_t(v)) c_{s,t}(F_s(x), v) dv. \quad (16)$$

This selected kernel is a regular conditional-law version for X , and a corresponding transition-density version is

$$p_{s,t}^X(x, y) = c_{s,t}(F_s(x), F_t(y)) p_t(y), \quad x \in E_s^\circ, \quad y \in E_t^\circ, \quad (17)$$

for every selected source state x and Lebesgue-a.e. target state y . As a conditional law, (16) is unique only for $p_s(x) dx$ -a.e. x . The exact quantifiers for its density identities are

$$\begin{aligned} \int_{E_t^\circ} p_{s,t}^X(x, y) dy &= 1, & \text{every } x \in E_s^\circ, \\ \int_{E_s^\circ} p_s(x) p_{s,t}^X(x, y) dx &= p_t(y), & \text{Lebesgue-a.e. } y, \\ \int_{E_r^\circ} p_{s,r}^X(x, z) p_{r,t}^X(z, y) dz &= p_{s,t}^X(x, y), & s < r < t, \text{ every } x \in E_s^\circ, \\ & & \text{Lebesgue-a.e. } y \in E_t^\circ. \end{aligned} \quad (18)$$

Equivalently, the Chapman–Kolmogorov identity holds for K^X at every selected x and every Borel target set.

Proof. The selected conditional law of U_t given $U_s = F_s(x)$ is $K_{s,t}(F_s(x), \cdot)$, so its pushforward by Q_t is (16). The change of variables $v = F_t(y)$ gives (17). Row normalization gives the first identity in (18); column normalization gives marginal propagation; and the substitution $w = F_r(z)$ in (3) gives its density Chapman–Kolmogorov identity. Integrating the latter in y gives the kernel identity for every Borel set. $\square$

2.2.3 Generators and the true-martingale qualification

For bounded continuous g , use the transition operators (7). The generator, martingale-problem, and localization conventions follow Ethier and Kurtz (1986, Chapter 4); the semimartingale and class-(D) facts used below follow Protter (2005, Chapters I–III).

Definition 2.9 (Right and extended generators). *For a Feller evolution, the right generator at time t is*

$$\mathcal{A}_t g = \lim_{h \downarrow 0} \frac{P_{t,t+h} g - g}{h} \quad (19)$$

whenever the limit exists in the chosen Banach-space norm. An unbounded time-dependent quantile $q(t, u)$ is said to lie in the time–space extended generator domain if a localized Dynkin formula holds with drift $(\partial_t + \mathcal{A}_t)q$ and the required integrability is finite.

Theorem 2.10 (Generator martingale theorem). *Let $q(t, u) = Q_t(u)$ lie in the time–space extended generator domain. Suppose that, for every $T < \infty$, the stopped family*

$$\{q(\tau, U_\tau) : \tau \text{ is a stopping time with } 0 \leq \tau \leq T\}$$

is uniformly integrable, that is, $q(\cdot, U_\cdot)$ is of class-(D) on every finite horizon. Then $X_t = q(t, U_t)$ is a true martingale if and only if

$$(\partial_t + \mathcal{A}_t)q(t, u) = 0 \quad \text{for } dt \otimes \text{Leb}(du)\text{-a.e. } (t, u). \quad (20)$$

Without the class-(D) or an equivalent uniform-integrability condition, the zero-drift equation generally establishes only a local martingale.

Proof. The extended Dynkin formula makes $q(t, U_t) - q(0, U_0) - \int_0^t (\partial_s + \mathcal{A}_s)q(s, U_s) ds$ a local martingale. Equation (20) therefore gives a local martingale; class-(D) upgrades it to a true martingale. Conversely, the finite-variation part of a special-semimartingale decomposition is unique, so a true martingale in the stated domain has zero drift almost everywhere. The measure-theoretic details, including the passage from $dt \otimes \mathbb{P}$ to $dt \otimes \text{Leb}$, are given in Appendix A.5. $\square$

2.2.4 Gaussian and diffusion copula generators

Let Φ and ϕ be the standard Normal CDF and density.

Proposition 2.11 (Brownian and deterministic-time-change generators). *For the uniformized Brownian family and $0 < s < t$, the pair-copula is Gaussian with correlation $\rho_{s,t} = \sqrt{s/t} \in (0, 1)$ and density*

$$c_{s,t}^G(u, v) = \frac{\phi_{\rho_{s,t}}(\Phi^{-1}(u), \Phi^{-1}(v))}{\phi(\Phi^{-1}(u)) \phi(\Phi^{-1}(v))}, \quad (21)$$

where ϕ_ρ is the standard bivariate Normal density with correlation ρ . The raw standardized Brownian example is defined on $(0, \infty)$. For $t > 0$, its local generator on $C_c^2(I)$ is

$$\mathcal{A}_t g(u) = -\frac{z\phi(z)}{t} g'(u) + \frac{\phi(z)^2}{2t} g''(u), \quad z = \Phi^{-1}(u). \quad (22)$$

If Brownian time is replaced by a differentiable, strictly increasing h with $h(t) > 0$, replace $\rho_{s,t}$ in (21) by $\sqrt{h(s)/h(t)} \in (0, 1)$, and

$$\mathcal{A}_t^h g(u) = \frac{h'(t)}{h(t)} \left[-z\phi(z)g'(u) + \frac{1}{2}\phi(z)^2 g''(u) \right]. \quad (23)$$

In particular, $h(t) = e^{2at}$ with $a > 0$ gives

$$\mathcal{A}g(u) = a\phi(z)^2 g''(u) - 2az\phi(z)g'(u). \quad (24)$$

For both clocks, $\rho_{s,t} = \rho_{s,r}\rho_{r,t}$ for $s < r < t$; Gaussian conditioning therefore gives the density Chapman–Kolmogorov identity (3). A specification whose time index includes $t = 0$ must use a clock with $h(0) > 0$ rather than $B_t/\sqrt{t}$ at the origin.

Proof. For $Z_t = B_t/\sqrt{t}$, Itô's formula gives $dZ_t = -Z_t(2t)^{-1}dt + t^{-1/2}dB_t$. Applying Itô again to $U_t = \Phi(Z_t)$ yields (22). The deterministic time change multiplies the local clock by $h'(t)$ and evaluates the Brownian coefficient at $h(t)$, which gives (23). $\square$

Proposition 2.12 (Uniformized diffusion generator). *Let*

$$dR_t = b(t, R_t)dt + \sqrt{a(t, R_t)}dW_t \quad (25)$$

take values in an interval (ℓ, r) . Suppose that its CDF $F \in C^{1,2}$ in the interior, $p = \partial_x F > 0$, and a is locally C^1 in x . Write $F_t^R(x) = F(t, x)$ and $Q_t^R = (F_t^R)^{-1}$. Assume that, for every $s < t$, selected kernels

$$K_{s,t}^R(x, B) = \int_B p_{s,t}^R(x, y) dy \quad (26)$$

are jointly Borel in (s, t, x) , form a Markov evolution for every $x \in (\ell, r)$ and Borel B , and give the entire regular conditional law, with no additional boundary atom. Then a selected version of its pair-copula density is

$$c_{s,t}(u, v) = \frac{p_{s,t}^R(Q_s^R(u), Q_t^R(v))}{p(t, Q_t^R(v))}. \quad (27)$$

for $(u, v) \in I^2$. As a Radon–Nikodym derivative it is unique only Lebesgue²-a.e.; the displayed formula specifies the version used by the transition kernel.

Assume additionally that the marginal density and coefficients have enough interior regularity for the classical forward equation $p_t = -\partial_x(bp) + \frac{1}{2}\partial_{xx}(ap)$, and that the left-boundary probability flux satisfies

$$b(t, x)p(t, x) - \frac{1}{2}\partial_x\{a(t, x)p(t, x)\} \longrightarrow 0 \quad \text{as } x \downarrow \ell \quad (28)$$

vanishes, then the local generator of $U_t = F_t(R_t)$ is

$$\mathcal{A}_t g(u) = [ap_x + \frac{1}{2}a_x p]_{x=Q_t^R(u)} g'(u) + \frac{1}{2}[ap^2]_{x=Q_t^R(u)} g''(u). \quad (29)$$

Proof. The conditional change of variables $y = Q_t^R(v)$ gives (27). Its row normalization is the normalization of the selected kernel (26); its column normalization holds for a.e. target rank and follows from marginal propagation $p(t, y) = \int_\ell^r p(s, x)p_{s,t}^R(x, y)dx$. With $z = Q_r^R(w)$ and $dw = p(r, z)dz$, the physical transition-density Chapman–Kolmogorov identity gives (3) for every selected source rank and a.e. target rank. Apply the time-dependent Itô formula to $F_t^R(R_t)$ and integrate the forward equation from ℓ to x . Condition (28) removes the boundary constant. A final Itô application to $g(U_t)$ gives (29). The Itô and diffusion-forward-equation results used here are standard; see Karatzas and Shreve (1991, Chapters 5–6). $\square$

Remark 2.13. The preceding interior result covers OU directly. CIR is degenerate at zero and must be treated on $(0, \infty)$ with its nonnegative solution, interior density, and boundary behavior stated separately; a uniformly elliptic whole-line assumption cannot be invoked for CIR.

2.3 Correct notions of mean reversion

2.3.1 Local, conditional, and auxiliary notions

Definition 2.14 (Forward local drift). *For an integrable real-valued Markov process with transition kernel $P_{s,t}(x, dy)$, its forward local drift is*

$$b_X(t, x) = \lim_{h \downarrow 0} \frac{\int y P_{t,t+h}(x, dy) - x}{h}, \quad (30)$$

whenever a finite limit exists. The definition is made for a chosen regular kernel version and for F_t -almost every $x \in E_t$; a continuous version may be extended to all interior states.

Definition 2.15 (One-dimensional Markov mean reversion). *An integrable real-valued Markov process X is locally mean reverting toward m if*

$$(i) \lim_{t \rightarrow \infty} \mathbb{E}[X_t] = m;$$

(ii) the forward drift (30) exists and

$$(x - m)b_X(t, x) < 0 \quad (x \neq m), \quad b_X(t, m) = 0 \quad (31)$$

whenever the indicated states belong to the support.

The quantity b_X is a drift, not a “reversion rate.”

Definition 2.16 (Strong conditional-mean reversion). *An integrable adapted process X has strong conditional-mean reversion toward m if a deterministic function $\mathcal{R}_{s,t} > 0$ exists such that*

$$\mathbb{E}[X_t - m \mid \mathcal{F}_s] = \mathcal{R}_{s,t}(X_s - m), \quad 0 \leq s < t, \quad (32)$$

$$\mathcal{R}_{s,r}\mathcal{R}_{r,t} = \mathcal{R}_{s,t}, \quad \lim_{t \rightarrow \infty} \mathcal{R}_{s,t} = 0. \quad (33)$$

A principal case is $\mathcal{R}_{s,t} = \exp\{\int_s^t \gamma(u) du\}$ with continuous $\gamma(u) < 0$.

Definition 2.17 (Finite-horizon and directional classes). *For model diagnostics we use two weaker labels, neither of which is silently called global mean reversion.*

(i) A finite-horizon conditional-reversion model satisfies (32) and is a one-dimensional Markov model only on a stated window $0 \leq t \leq T$.

(ii) A directional-reversion model has a drift pointing toward a threshold c , but c has not been proved equal to the limiting mean.

An observed projection of a Markov vector is called a latent-state extension; it is not asserted to be a one-dimensional Markov process.

Proposition 2.18 (Strong reversion implies local reversion). *Suppose that X is an integrable Markov process and that Definition 2.16 holds with $\mathcal{R}_{s,t} = \exp\{\int_s^t \gamma(u) du\}$, where γ is continuous and negative. Fix t . Select a regular transition-kernel version for which, on one F_t -full set D_t , the conditional-mean identity holds for every sufficiently small $h > 0$ (it is enough that it holds for rational $h \downarrow 0$ and that the kernel first moment is right-continuous in h). Then, for F_t -almost every $x \in D_t$,*

$$b_X(t, x) = \gamma(t)(x - m). \quad (34)$$

If the two sides have continuous versions in x , the identity extends to all interior support states. Moreover, for any integrable X_s , $\mathbb{E}[X_t] \rightarrow m$. Thus the selected version satisfies Definition 2.15 with its natural almost-everywhere interpretation.

Proof. The Markov property and the selected common kernel version give, for $x \in D_t$,

$$\int y P_{t,t+h}(x, dy) - x = \{e^{\int_t^{t+h} \gamma(a) da} - 1\}(x - m).$$

Divide by h and let $h \downarrow 0$. Continuity of γ gives (34); this argument also records precisely why separate null sets for uncountably many h would not suffice. Taking expectations in (32) gives $\mathbb{E}[X_t] - m = \mathcal{R}_{s,t}\{\mathbb{E}[X_s] - m\} \rightarrow 0$. Since $\gamma(t) < 0$, the local sign condition follows. $\square$

Remark 2.19. The converse is false in general: an instantaneous sign condition and an unconditional limiting mean do not by themselves determine every finite-horizon conditional mean. This is why the local and strong definitions are kept separate.

Proposition 2.20 (Martingale rescaling). *Let X be an integrable Markov process satisfying Definition 2.16. Then*

$$M_t = \frac{X_t - m}{\mathcal{R}_{0,t}} \quad (35)$$

is a Markov martingale. Conversely, if M is an integrable Markov martingale, $\mathcal{R}_{0,t} > 0$ is deterministic and multiplicative, and $X_t = m + \mathcal{R}_{0,t}M_t$, then X satisfies (32). The two processes have the same pair-copulas. If H_t is the marginal CDF of M_t , their quantiles satisfy

$$H_t^{-1}(u) = \frac{Q_t(u) - m}{\mathcal{R}_{0,t}}. \quad (36)$$

When the hypotheses of Theorem 2.10 hold, this is equivalent to

$$(\partial_t + \mathcal{A}_t)Q(t, u) - \gamma(t)\{Q_t(u) - m\} = 0. \quad (37)$$

Proof. Multiplicativity gives $\mathcal{R}_{0,t} = \mathcal{R}_{0,s}\mathcal{R}_{s,t}$. Hence

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = \frac{\mathcal{R}_{s,t}(X_s - m)}{\mathcal{R}_{0,t}} = M_s.$$

The converse follows by reversing this calculation. At each time the map between X_t and M_t is strictly increasing and affine, so it preserves pair-copulas and gives (36). Applying the generator martingale theorem to that quantile and differentiating $\mathcal{R}_{0,t}$ gives (37). $\square$

2.3.2 Exact copula and generator characterizations

Theorem 2.21 (Exact copula criterion for strong mean reversion). *Under Assumption 2.6, X satisfies (32) if and only if*

$$\int_I \{Q_t(v) - m\} c_{s,t}(u, v) dv = \mathcal{R}_{s,t}\{Q_s(u) - m\} \quad (38)$$

for each fixed $s < t$ and for Leb-almost every u ; the exceptional set may depend on (s, t) . If both sides admit continuous versions in u , the equality extends to every $u \in I$.

Proof. By (14), the left side of (38), evaluated at $u = U_s$, equals $\mathbb{E}[X_t - m \mid \mathcal{F}_s]$. Its right side then equals $\mathcal{R}_{s,t}(X_s - m)$. Since U_s is uniform, equality almost surely is equivalent to the displayed density identity for Leb-almost every u . $\square$

Proposition 2.22 (Generator sign criterion). *Suppose $Q(t, u)$ is in the time-space extended generator domain. Fix t and an interior rank u at which the integral generator limit (124) exists. For $r_h(v) = \{Q(t+h, v) - Q(t, v)\}/h$ and $g(v) = \partial_t Q(t, v)$, assume either the bounded uniform*

conditions (117) and (119), with g bounded and continuous at u , or the weighted conditions (121)–(123). Then

$$b_X(t, Q_t(u)) = (\partial_t + \mathcal{A}_t)Q(t, u). \quad (39)$$

Consequently, after separately imposing the limiting-mean condition, Definition 2.15 is equivalent to

$$\text{sgn}\{(\partial_t + \mathcal{A}_t)Q(t, u)\} = \text{sgn}\{m - Q_t(u)\} \quad (40)$$

for almost every interior u , with zero drift at $Q_t(u) = m$.

Proof. Insert $X_t = Q_t(U_t)$ into Definition 2.14 and use the definition of the time-space extended generator. Strict monotonicity converts the physical-state sign condition into (40). A complete kernel-integral proof, with separate bounded and weighted conditions for the moving time difference quotient, is given in Appendix A.3. $\square$

2.3.3 Stationary Gaussian copula and Gamma-margin diagnostics

Let Z be the stationary standardized OU process

$$dZ_t = -aZ_t dt + \sqrt{2a} dW_t, \quad a > 0, \quad Z_t \sim \mathcal{N}(0, 1). \quad (41)$$

Its copula is Gaussian with correlation $e^{-a(t-s)}$.

Proposition 2.23 (Stationary Gaussian-copula sign criterion). *Let F have a finite mean m , let $g(z) = F^{-1}(\Phi(z))$, and suppose g is strictly increasing, C^2 , and in the extended generator domain of (41). Let z_F be the unique number satisfying $g(z_F) = m$. If*

$$\text{sgn}\{g''(z) - zg'(z)\} = \text{sgn}(z_F - z) \quad \text{for every } z \in \mathbb{R}, \quad (42)$$

then $X_t = g(Z_t)$ is locally mean reverting toward m .

Proof. Itô's formula gives local drift $a\{g''(z) - zg'(z)\}$. Since g is strictly increasing, $\text{sgn}(m - g(z)) = \text{sgn}(z_F - z)$. Stationarity gives $\mathbb{E}[X_t] = m$ at every time. Integrability and the extended-domain assumption justify the drift calculation. $\square$

Example 2.24 (Stationary Gamma margin: a necessary root is not sufficient). Let G_ν be the quantile of $\text{Gamma}(\nu, 1)$, with $\nu > 0$, and set $Q(u) = sG_\nu(u)$, $s > 0$. Define

$$u_\nu = F_{\nu,1}(\nu), \quad z_\nu = \Phi^{-1}(u_\nu). \quad (43)$$

At the target mean $m = \nu s$, the zero-drift requirement reduces to

$$\Gamma(\nu) \left(\frac{e}{\nu}\right)^\nu \phi(z_\nu) - 2z_\nu = 0. \quad (44)$$

Equation (44) is only necessary. The model is locally mean reverting only if the full condition

$$\text{sgn}\{\mathcal{A}Q(u)\} = \text{sgn}\{\nu s - Q(u)\} \quad \text{for almost every } u \in I \quad (45)$$

also holds, with the required endpoint integrability. No existence claim is made from the scalar root alone.

Example 2.25 (Time-varying Gamma scale). Let $Q_t(u) = s(t)G_\nu(u)$, where $s(t) > 0$ is differentiable and $s(t) \rightarrow s_\infty \in (0, \infty)$. Write $z = \Phi^{-1}(u)$ and

$$\mathcal{G}_\nu(u) = 2\phi(z)zG'_\nu(u) - \phi(z)^2G''_\nu(u). \quad (46)$$

For the Gaussian-copula generator (24), the quantile drift is

$$D_t(u) = s'(t)G_\nu(u) - as(t)\mathcal{G}_\nu(u). \quad (47)$$

The correct target is $m = \nu s_\infty$, and the necessary and sufficient local sign test is

$$\operatorname{sgn} D_t(u) = \operatorname{sgn}\{m - s(t)G_\nu(u)\} \quad \text{for almost every } u \in I. \quad (48)$$

If u_t solves $s(t)G_\nu(u_t) = m$, then

$$\frac{s'(t)}{s(t)} = a \frac{\mathcal{G}_\nu(u_t)}{G_\nu(u_t)} \quad (49)$$

is a necessary zero-point equation, not a sufficient global condition. Its limit also inherits the stationary root requirement (44).

3 Mean-Reverting Processes from Monotone Diffusion Transforms

3.1 Construction theorem and observable density

Let R be a Markov diffusion on an interval E ,

$$dR_t = b(R_t) dt + \sigma(R_t) dW_t, \quad \mathcal{L}h = bh_x + \frac{1}{2}\sigma^2 h_{xx}. \quad (50)$$

Theorem 3.1 (Strict diffusion-transform theorem). *Let $f \in C^{1,2}([0, \infty) \times E)$ and suppose that $x \mapsto f(t, x)$ is strictly increasing for every relevant t . Let γ be continuous and let*

$$\mathcal{R}_{s,t} = \exp\left\{\int_s^t \gamma(u) du\right\}. \quad (51)$$

Assume that $\mathbb{E}|f(0, R_0)| < \infty$ and

$$(\partial_t + \mathcal{L})f(t, x) = \gamma(t)\{f(t, x) - m\} \quad (52)$$

and, for every $T < \infty$,

$$\mathbb{E} \int_0^T \mathcal{R}_{0,t}^{-2} \sigma(R_t)^2 f_x(t, R_t)^2 dt < \infty. \quad (53)$$

Then $Y_t = f(t, R_t)$ is Markov and

$$\mathbb{E}[Y_t - m \mid \mathcal{F}_s] = \mathcal{R}_{s,t}(Y_s - m). \quad (54)$$

If $\gamma(t) < 0$, $\mathcal{R}_{s,t} \rightarrow 0$, and the initial first moment is finite, then Y has strong and local mean reversion toward m .

Conversely, if the strictly monotone $C^{1,2}$ transform satisfies (54) and its semimartingale decomposition is integrable, uniqueness of the drift part yields

$$(\partial_t + \mathcal{L})f(t, R_t) = \gamma(t)\{f(t, R_t) - m\} \quad \text{for } dt \otimes \mathbb{P}\text{-a.e. } (t, \omega). \quad (55)$$

Equivalently, the PDE residual is zero for $dt \otimes \operatorname{Law}(R_t)(dx)$ -almost every (t, x) . It upgrades to the pointwise PDE (52) on $[0, \infty) \times E$ if, for example, the residual is continuous and $\operatorname{Law}(R_t)$ has full support in E for Lebesgue-a.e. t (with time continuity filling the remaining times).

Proof. Itô's formula and (52) give

$$d(Y_t - m) = \gamma(t)(Y_t - m) dt + \sigma(R_t) f_x(t, R_t) dW_t.$$

Multiplying by $\mathcal{R}_{0,t}^{-1}$ removes the drift. Condition (53) makes the resulting stochastic integral a true square-integrable martingale on every finite horizon, proving (54). Strict invertibility transfers the Markov property from R to Y . The converse follows from uniqueness of the finite-variation part in the almost-everywhere sense stated above. These Itô-integral and semimartingale-uniqueness steps are covered, for example, by Karatzas and Shreve (1991) and Protter (2005). $\square$

Proposition 3.2 (Transition density under a monotone transform). *Suppose a selected transition kernel of R has density $p_{s,t}(x_s, x_t)$ for every source state x_s , and the observations y_j lie in $f(j, E)$ with unique inverse states $x_j = f^{-1}(j, y_j)$ for $j = s, t$. Then*

$$q_{s,t}(y_s, y_t) = p_{s,t}(x_s, x_t) \left| \frac{\partial f^{-1}(t, y_t)}{\partial y_t} \right| = \frac{p_{s,t}(x_s, x_t)}{|f_x(t, x_t)|}. \quad (56)$$

for every selected source observation y_s and Lebesgue-a.e. target observation y_t . This is a valid interior density only where the support, uniqueness, and Jacobian conditions hold. For the strictly increasing transforms used here, the probability ranks are unchanged. Hence the pair-copula CDFs satisfy the pointwise identity

$$C_{s,t}^Y(u, v) = C_{s,t}^R(u, v), \quad (u, v) \in [0, 1]^2, \quad (57)$$

and their Radon–Nikodym densities satisfy

$$c_{s,t}^Y(u, v) = c_{s,t}^R(u, v) \quad \text{for Lebesgue-a.e. } (u, v). \quad (58)$$

One may, and in the empirical likelihood does, select the same pointwise density version on both sides. The copula invariance in (57) is the standard invariance under strictly increasing marginal transformations; see Nelsen (2006, Chapter 2).

Proof. Condition on the uniquely recovered state $R_s = x_s$ and apply the one-dimensional change-of-variables theorem at time t . Moreover, $F_t^Y(f(t, x)) = F_t^R(x)$ for every interior x . Substitution into the joint CDF proves (57); uniqueness of Radon–Nikodym derivatives proves (58). $\square$

3.2 OU base process

Use the centered OU process

$$dR_t = -\kappa R_t dt + \sigma dW_t, \quad \kappa > 0, \quad \sigma > 0. \quad (59)$$

For $\Delta = t - s$,

$$R_t \mid R_s = x \sim \mathcal{N}\left(e^{-\kappa\Delta}x, \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa\Delta})\right). \quad (60)$$

The noncentered OU is obtained by replacing R_t with $R_t - \bar{r}$. It therefore satisfies

$$\mathbb{E}[R_t - \bar{r} \mid \mathcal{F}_s] = e^{-\kappa(t-s)}(R_s - \bar{r}), \quad \lim_{t \rightarrow \infty} \mathbb{E}[R_t \mid \mathcal{F}_s] = \bar{r}. \quad (61)$$

Set $\gamma = -d$ with $d > 0$. The transform PDE is

$$f_t + \frac{1}{2}\sigma^2 f_{xx} - \kappa x f_x + d(f - m) = 0. \quad (62)$$

Example 3.3 (Affine transformed OU). For $A_0 > 0$ and $B_0 \in \mathbb{R}$,

$$f(t, x) = m + A_0 e^{(\kappa-d)t} x + B_0 e^{-dt} \quad (63)$$

solves (62) and is strictly increasing on $\mathbb{R}$. If the base OU is stationary, then

$$Y_t \sim \mathcal{N}\left(m + B_0 e^{-dt}, \frac{A_0^2 \sigma^2}{2\kappa} e^{2(\kappa-d)t}\right). \quad (64)$$

The variance may grow when $\kappa > d$; this does not contradict the first-moment identity (54), but the transformed process is then not stationary.

Proof. Substitution leaves the coefficient equations $A' + (d - \kappa)A = 0$ and $B' + dB = 0$. The derivative $f_x = A_0 e^{(\kappa-d)t}$ is positive. The marginal law follows from the affine Normal transformation. $\square$

Example 3.4 (Exponential transformed OU). Let

$$f(t, x) = m + e^{A(t)x+B(t)} + C(t), \quad (65)$$

$$A(t) = A_0 e^{\kappa t}, \quad (66)$$

$$B(t) = B_0 - \frac{\sigma^2 A_0^2}{4\kappa} (e^{2\kappa t} - 1) - dt, \quad (67)$$

$$C(t) = C_0 e^{-dt}. \quad (68)$$

If $A_0 > 0$, this is strictly increasing and its support is $(m + C(t), \infty)$. It is positive-valued when $m + C(t) \geq 0$. For stationary R , the shifted variable $Y_t - m - C(t)$ is lognormal with log mean $B(t)$ and log variance $A(t)^2 \sigma^2 / (2\kappa)$.

Proof. Substitution in (62) gives

$$A' = \kappa A, \quad B' = -\frac{1}{2}\sigma^2 A^2 - d, \quad C' = -dC,$$

whose solution is (66)–(68). The derivative $f_x = A e^{Ax+B}$ is positive. Gaussian exponential moments verify (53) on finite horizons. $\square$

Proposition 3.5 (OU Kummer candidate is not an admissible monotone model). *Let $J(p, q, z)$ solve $zJ_{zz} + (q - z)J_z - pJ = 0$. The separated family*

$$f(t, x) = m + e^{-\zeta t} J\left(\frac{\zeta - d}{2\kappa}, \frac{1}{2}, \frac{\kappa x^2}{\sigma^2}\right) \quad (69)$$

solves (62) wherever it is smooth. Every nonconstant member is, however, noninjective on the OU state space $\mathbb{R}$ and therefore does not satisfy Theorem 3.1 or Proposition 3.2.

Proof. Direct substitution gives the stated Kummer equation. The spatial dependence is through x^2 , so $f(t, x) = f(t, -x)$ for every x . A nonconstant even function cannot be one-to-one on $\mathbb{R}$. $\square$

Remark 3.6. A density formed by summing two inverse branches would define a different observation model. The transformation would no longer preserve the base copula, and the observed scalar process need not be Markov; this is outside the present one-dimensional monotone framework.

3.3 CIR base process

Let

$$dR_t = \kappa(\bar{r} - R_t)dt + \sigma\sqrt{R_t}dW_t, \quad \kappa > 0, \quad \bar{r} > 0, \quad \sigma > 0, \quad (70)$$

with state space $[0, \infty)$. The Feller condition $2\kappa\bar{r} \geq \sigma^2$ is a convenient sufficient condition for zero to be unattainable when the initial state is positive. For $\Delta = t - s$, the exact transition law is

$$R_t \mid R_s = x \stackrel{d}{=} c_\Delta \chi_\nu'^2(\lambda_\Delta), \quad (71)$$

$$c_\Delta = \frac{\sigma^2(1 - e^{-\kappa\Delta})}{4\kappa}, \quad \nu = \frac{4\kappa\bar{r}}{\sigma^2}, \quad \lambda_\Delta = \frac{4\kappa e^{-\kappa\Delta}x}{\sigma^2(1 - e^{-\kappa\Delta})}. \quad (72)$$

The conditional mean has the same linear reversion form as OU:

$$\mathbb{E}[R_t - \bar{r} \mid \mathcal{F}_s] = e^{-\kappa(t-s)}(R_s - \bar{r}). \quad (73)$$

When started from its invariant law,

$$R_t \sim \text{Gamma}\left(\frac{2\kappa\bar{r}}{\sigma^2}, \frac{\sigma^2}{2\kappa}\right), \quad (74)$$

where the second parameter is scale. With $\gamma = -d$, the corrected transform PDE is

$$f_t + \frac{1}{2}\sigma^2 x f_{xx} + \kappa(\bar{r} - x)f_x + d(f - m) = 0. \quad (75)$$

The boundary statement, noncentral chi-square transition law, conditional mean, and invariant Gamma law are the standard CIR results; see Cox et al. (1985).

Example 3.7 (Affine transformed CIR). For $A_0 > 0$ and $B_0 \in \mathbb{R}$, define

$$A(t) = A_0 e^{(\kappa-d)t}, \quad (76)$$

$$B(t) = m - A_0 \bar{r} e^{(\kappa-d)t} + B_0 e^{-dt}, \quad (77)$$

$$f(t, x) = A(t)x + B(t). \quad (78)$$

Then f solves (75), is strictly increasing on the CIR state space, and has support $[B(t), \infty)$. The original manuscript's formula is the special case $m = \bar{r}$ after a harmless reparameterization of B_0 .

Proof. The affine coefficient equations are $A' + (d - \kappa)A = 0$ and $B' + d(B - m) + \kappa\bar{r}A = 0$. Equations (76)–(77) solve them, and $f_x = A > 0$. Polynomial moments of CIR verify the finite-horizon martingale condition. $\square$

Proposition 3.8 (Quadratic CIR transform: exact finite-horizon condition). *Let*

$$f(t, x) = A(t)x^2 + B(t)x + C(t), \quad A_0 > 0, \quad (79)$$

and set

$$\eta = \frac{A_0(\sigma^2 + 2\kappa\bar{r})}{\kappa}. \quad (80)$$

The coefficient solution is

$$A(t) = A_0 e^{(2\kappa-d)t}, \quad (81)$$

$$B(t) = e^{(\kappa-d)t} \{B_0 - \eta(e^{\kappa t} - 1)\}, \quad (82)$$

$$\begin{aligned} C(t) = m + \frac{\bar{r}\eta}{2} e^{(2\kappa-d)t} - \bar{r}(B_0 + \eta) e^{(\kappa-d)t} \\ + \left(C_0 - m + \bar{r}B_0 + \frac{\bar{r}\eta}{2}\right) e^{-dt}. \end{aligned} \quad (83)$$

It solves (75). On a fixed window $[0, T]$, the map $x \mapsto f(t, x)$ is increasing on $[0, \infty)$ for every $t \leq T$ if and only if

$$B_0 \geq \eta(e^{\kappa T} - 1). \quad (84)$$

Under a strict inequality the interior inverse Jacobian is strictly finite. No finite B_0 makes a nondegenerate quadratic transform globally increasing for all $t \geq 0$ under the standard CIR parameter restrictions.

Proof. Matching the coefficients of $x^2, x, 1$ gives

$$\begin{aligned} A' + (d - 2\kappa)A &= 0, \\ B' + (d - \kappa)B + (\sigma^2 + 2\kappa\bar{r})A &= 0, \\ C' + d(C - m) + \kappa\bar{r}B &= 0, \end{aligned}$$

which yields (81)–(83). Since $f_x = 2A(t)x + B(t)$ and $A(t) > 0$, monotonicity on the nonnegative half-line is equivalent to $B(t) \geq 0$. Formula (82) is nonnegative for all $t \leq T$ exactly under (84). Its bracket tends to $-\infty$ as $t \rightarrow \infty$, proving the global impossibility. $\square$

Remark 3.9 (Why $\kappa < 0$ is not the same CIR model). Algebraically, when $\kappa < 0$ the quadratic map can be globally increasing if $A_0 > 0$ and $\min\{B_0, B_0 + \eta\} \geq 0$. This does not rescue the standard CIR model. If $\bar{r} > 0$, the boundary drift $\kappa\bar{r} < 0$ points out of the nonnegative state space. If $\bar{r} < 0$ is chosen to make the intercept positive, the drift is explosive and there is no invariant Gamma law or mean reversion. Such a process would be a separately defined supercritical square-root diffusion, not a relaxation of (70).

Corollary 3.10 (Quadratic inverse and density). *Under the strict finite-window condition and for an interior observation $y > C(t)$, the unique nonnegative inverse is*

$$x = \frac{2\{y - C(t)\}}{B(t) + \sqrt{B(t)^2 + 4A(t)\{y - C(t)\}}}, \quad \left| \frac{\partial x}{\partial y} \right| = \frac{1}{2A(t)x + B(t)}. \quad (85)$$

Together with (71), this gives the finite-horizon observation density through (56).

Example 3.11 (Exponential transformed CIR). Let

$$f(t, x) = m + e^{A(t)x + B(t)} + D(t), \quad (86)$$

$$A(t) = \frac{2\kappa}{\sigma^2 + (2\kappa/A_0 - \sigma^2)e^{-\kappa t}}, \quad (87)$$

$$\begin{aligned} B(t) &= -dt - \frac{2\kappa\bar{r}}{\sigma^2} \log[(2\kappa/A_0 - \sigma^2) + \sigma^2 e^{\kappa t}] \\ &\quad + B_0 + \frac{2\kappa\bar{r}}{\sigma^2} \log(2\kappa/A_0), \end{aligned} \quad (88)$$

$$D(t) = D_0 e^{-dt}. \quad (89)$$

For $A_0 > 0$, the logarithm arguments and denominator are positive and the map is strictly increasing. Its support is

$$[m + D(t) + e^{B(t)}, \infty). \quad (90)$$

If the base CIR is stationary, a sufficient and necessary initial exponential moment restriction is

$$0 < A_0 < \frac{2\kappa}{\sigma^2}. \quad (91)$$

Under these conditions the positive local-martingale component has constant expectation and is therefore a true martingale; this verifies the conditional mean identity even when the square-integrability criterion (53) is stronger than needed.

Proof. The coefficient equations are

$$A' + \frac{1}{2}\sigma^2 A^2 - \kappa A = 0, \quad B' + d + \kappa\bar{r}A = 0, \quad D' + dD = 0.$$

Their solution is (87)–(89). Since $R_t \geq 0$, the minimum of the exponential term is $e^{B(t)}$, proving (90). The stationary Gamma MGF is finite exactly below rate $2\kappa/\sigma^2$, which yields (91). More explicitly, put $p_0 = \sigma^2 A_0/(2\kappa) \in (0, 1)$ and $q = 2\kappa\bar{r}/\sigma^2$. Then

$$B(t) = B_0 - dt - q \log(1 - p_0 + p_0 e^{\kappa t}), \quad 1 - \frac{\sigma^2 A(t)}{2\kappa} = \frac{1 - p_0}{1 - p_0 + p_0 e^{\kappa t}}.$$

Using the stationary Gamma MGF shows

$$\mathbb{E} \exp\{dt + A(t)R_t + B(t)\} = e^{B_0}(1 - p_0)^{-q},$$

which is constant. The nonnegative local-martingale component is therefore a true martingale. $\square$

3.4 CIR special functions and the corrected reduction

Let $M(p, q, z) = {}_1F_1(p; q; z)$ denote Kummer's function and $U(p, q, z)$ Tricomi's function, with limiting definitions used at exceptional integer parameters. We use the definitions, derivative identities, and asymptotic conventions of Olver et al. (2010, Chapter 13).

Proposition 3.12 (CIR Kummer separated family). *Set*

$$p = \frac{\zeta - d}{\kappa}, \quad q = \frac{2\kappa\bar{r}}{\sigma^2}, \quad z = \frac{2\kappa x}{\sigma^2}. \quad (92)$$

If

$$J(p, q, z) = C_1 M(p, q, z) + C_2 U(p, q, z) \quad (93)$$

is a well-defined branch satisfying the relevant boundary conditions, then

$$f(t, x) = m + e^{-\zeta t} J(p, q, z) \quad (94)$$

solves (75). It defines a model in the present framework only if the selected branch is strictly increasing on $[0, \infty)$, has a unique stable inverse on its image, and satisfies the moment condition in Theorem 3.1 for the stated initial law.

For example, $p, q > 0$, $C_1 > 0$, $C_2 = 0$ gives

$$\partial_z M(p, q, z) = \frac{p}{q} M(p+1, q+1, z) > 0, \quad (95)$$

but this branch is not integrable under the stationary Gamma law because of its e^z asymptotic term. It may be integrable at each finite time from a deterministic initial state. Thus the separated equation is correct, whereas an unspecified “special-function CIR model” is not a complete statistical model.

Proof. Substitution gives $zJ_{zz} + (q - z)J_z - pJ = 0$. The derivative identity and standard positive series for M prove monotonicity in the stated branch. The cited Kummer asymptotic is $M(p, q, z) \sim \Gamma(q)e^z z^{p-q}/\Gamma(p)$; multiplying by the invariant $z^{q-1}e^{-z}$ density leaves a nonintegrable z^{p-1} tail for $p > 0$. $\square$

Proposition 3.13 (Correction of a proposed CIR reduction). *The proposed exponential change of variables does not transform (75) into $u_\tau = zu_{zz}$; direct substitution has a nonzero residual for generic parameters. A correct direct reduction is*

$$f(t, x) - m = e^{-dt} u(\tau, z), \quad z = xe^{\kappa t}, \quad \tau = \frac{\sigma^2}{2\kappa}(1 - e^{\kappa t}) \leq 0, \quad (96)$$

which yields the following equation when τ is read as a reverse-time coordinate on the nonpositive half-line:

$$u_\tau = zu_{zz} + \frac{2\kappa\bar{r}}{\sigma^2} u_z. \quad (97)$$

Therefore the three functions listed after the proposed reduction are not thereby solutions of the original CIR transform PDE.

Proof. Differentiate (96) and substitute into (75); the κzu_z terms cancel and the remaining terms give (97). The coefficient of u_z is strictly positive under the standard CIR parameters and cannot generally be discarded. $\square$

4 Mean-Reverting Processes from Mixed Copula Families

4.1 Transition-density construction and preservation of mean reversion

Let $P_{s,t}$ be a Markov evolution on a state space E with common invariant probability law π ; thus the kernel Chapman–Kolmogorov identity holds for every selected source state x and every measurable target set. Assume that, for $s < t$, selected jointly measurable versions are dominated by a common σ -finite measure μ_0 ,

$$P_{s,t}(x, dy) = p_{s,t}(x, y)\mu_0(dy), \quad \pi(dy) = p_\pi(y)\mu_0(dy).$$

Here the first equality is imposed for every x as an equality of measures; the densities in y remain unique only μ_0 -a.e. Define the independence/reset operator

$$\Pi(x, A) = \pi(A), \quad (\Pi h)(x) = \int_E h(z)\pi(dz). \quad (98)$$

Let $\lambda : [0, \infty) \rightarrow [0, \infty)$ be right-continuous and locally bounded, and set

$$a_{s,t} = \exp\left\{-\int_s^t \lambda(u) du\right\}. \quad (99)$$

Theorem 4.1 (Consistent reset density mixture). *The kernels*

$$P_{s,t}^{\text{mix}} = a_{s,t}P_{s,t} + (1 - a_{s,t})\Pi \quad (100)$$

form a Markov evolution and preserve π . Their transition densities are

$$p_{s,t}^{\text{mix}}(x, y) = a_{s,t}p_{s,t}(x, y) + (1 - a_{s,t})p_\pi(y). \quad (101)$$

where the right-hand side defines the selected version; its density Chapman–Kolmogorov identity holds for every x and μ_0 -a.e. y , while the corresponding kernel identity holds for every x and every measurable target set. Assume additionally that $E \subseteq \mathbb{R}$, the process is started from π , the invariant CDF F_π is continuous and strictly increasing on the interior of its support, and the stationary base pair-copula has density $c_{s,t}$. Then the mixed pair-copula and its density are

$$C_{s,t}^{\text{mix}} = a_{s,t}C_{s,t} + (1 - a_{s,t})\Pi_C, \quad \Pi_C(u, v) = uv, \quad (102)$$

$$c_{s,t}^{\text{mix}}(u, v) = a_{s,t}c_{s,t}(u, v) + (1 - a_{s,t}). \quad (103)$$

The copula identity (102) is pointwise on $[0, 1]^2$. Equation (103) is a Lebesgue-a.e. density identity, and its right-hand side is the selected version used by the mixed kernel; with that selection, its density consistency holds for every source rank u and a.e. target rank v . If the base generator is $\mathcal{G}_t$, then, on any invariant generator domain whose functions are integrable under π ,

$$\mathcal{G}_t^{\text{mix}}h(x) = \mathcal{G}_th(x) + \lambda(t)\left\{\int_E h(z)\pi(dz) - h(x)\right\}. \quad (104)$$

Proof. The survival weights are multiplicative. Invariance gives $P\Pi = \Pi P = \Pi$ and $\Pi^2 = \Pi$. Expanding $P_{s,r}^{\text{mix}}P_{r,t}^{\text{mix}}$ therefore reduces it to (100) with weight $a_{s,r}a_{r,t} = a_{s,t}$. Taking densities gives (101); uniformization under stationary start gives (102) and the a.e. identity (103). Selecting its right-hand side as the density version, its margins equal one, and direct expansion using (2)–(3) gives

$$\int_I c_{s,r}^{\text{mix}}(u, w)c_{r,t}^{\text{mix}}(w, v)dw = a_{s,t}c_{s,t}(u, v) + (1 - a_{s,t}) = c_{s,t}^{\text{mix}}(u, v).$$

for every u and a.e. v ; integration in v gives exact kernel consistency for every target Borel set. The right derivative of (100) at $t = s+$ gives (104). $\square$

Theorem 4.2 (Exact conditional mean under stationary resets). *Suppose the base process has invariant mean $m = \int_E z\pi(dz)$ and satisfies*

$$\mathbb{E}[Z_t - m \mid Z_s = x] = \mathcal{R}_{s,t}(x - m). \quad (105)$$

Then the reset-mixture process satisfies

$$\mathbb{E}[Y_t - m \mid Y_s = x] = a_{s,t}\mathcal{R}_{s,t}(x - m). \quad (106)$$

Its local drift is

$$b_{\text{mix}}(t, x) = b_0(t, x) + \lambda(t)(m - x). \quad (107)$$

Thus a nonnegative reset intensity adds the local drift $\lambda(t)(m - x)$. Formula (106) is an exact identity; comparison of decay rates requires additional sign and parametrization conditions stated in the following corollary.

Proof. The base component of (100) contributes $a_{s,t}\mathcal{R}_{s,t}(x - m)$. A draw from π has centered mean zero, so the reset component contributes zero. This proves (106); its right derivative gives (107). $\square$

Corollary 4.3 (Strict acceleration under exponential persistence). *In addition to Theorem 4.2, suppose*

$$\mathcal{R}_{s,t} = \exp\left\{\int_s^t \gamma(u) du\right\}, \quad \gamma(u) < 0,$$

and $a_{s,t} = \exp\{-\int_s^t \lambda(u) du\}$ with $\lambda \geq 0$. Then

$$a_{s,t}\mathcal{R}_{s,t} = \exp\left\{\int_s^t [\gamma(u) - \lambda(u)] du\right\} \leq \mathcal{R}_{s,t}.$$

The inequality is strict exactly when $\int_s^t \lambda(u) du > 0$. Hence, on such an interval and for $x \neq m$, the absolute conditional-mean deviation is strictly smaller than under the base process.

Remark 4.4 (A precise theoretical advantage). The mixed model does not alter any one-time marginal under stationary start, but multiplies the base conditional-mean persistence by $a_{s,t}$. Hence $a_{s,t}\mathcal{R}_{s,t}$, the implied half-life, or the additive rate $-\partial_{t+} \log a_{s,t} = \lambda(t)$ are clear structural measures of its extra reversion channel. Whether this produces superior out-of-sample log scores or calibration is a separate empirical question.

4.2 Poisson realization

Stochastic resetting and restart have an established literature; see, for example, Evans and Majumdar (2011) and the review of Evans et al. (2020). The result below does not claim the reset mechanism itself as new. Its role is to realize the selected Markov mixture (100), and therefore its margin preservation, observable density, and conditional-mean identity, on one probability space.

Theorem 4.5 (Independent-copy reset construction). *Let $\{Z^{(j)} : j \geq 0\}$ be independent copies of a stationary Markov process with transition evolution P and invariant law π . Let N be an independent nonhomogeneous Poisson process with intensity $\lambda(t)$, and set*

$$Y_t = Z_t^{(N_t)}. \quad (108)$$

Then Y is Markov, stationary in its one-time marginals, and has transition kernel (100) and transition density (101). Under the stationary continuous-margin conditions of Theorem 4.1, it also has Copula density (103). A constant λ gives $a_{s,t} = e^{-\lambda(t-s)}$; a positive deterministic seasonal intensity is also admissible.

Proof. Conditional on no jump in $(s, t]$, whose probability is $a_{s,t}$, the same copy evolves with kernel $P_{s,t}$. Conditional on at least one jump, the active copy at t is independent of the history through time s and has law π . The resulting conditional law is (100), whose density is (101), and it is a Markov evolution by Theorem 4.1. $\square$

5 Empirical Evidence from Kraken USDT/USD

5.1 Data and evaluation design

The primary state is $X_t = \log P_t$, where P_t is the Kraken USDT/USD close. The baseline sample runs from January 2021 through December 2025. No prices are imputed. A transition enters the five-minute likelihood only when two observations are exactly 300 seconds apart; longer gaps start a new path segment. Estimation uses at most 120,000 systematically spaced exact pairs, and all available pairs up to the same cap are used for evaluation. The training period is 2021–2023, validation is 2024, and 2025 is retained as post-validation evidence. Margins and model parameters used outside the training period are never re-ranked or re-estimated on validation or test observations.

The full baseline contains an empirical mean of 0.6713 basis points in log-price units and a standard deviation of 7.3593 basis points. Its sample kurtosis is 286.9, driven by a small number of unusually large departures from the peg. This combination of a narrow center and extreme tails is precisely the setting in which conditional likelihood can prefer a contamination component that is not a credible invariant margin.

5.2 Central transition models

The OU benchmark is

$$dX_t = \kappa(\theta - X_t) dt + \sigma dW_t, \quad \tau = \frac{\sigma}{\sqrt{2\kappa}},$$

where τ is its invariant standard deviation. The mixed transition density implied by Section 4 is

$$p_h^{\text{mix}}(y | x) = e^{-\lambda h} p_h^{\text{OU}}(y | x) + (1 - e^{-\lambda h}) \phi(y; \theta, \tau^2), \quad (109)$$

$$\mathbb{E}[X_{t+h} - \theta | X_t = x] = e^{-(\kappa+\lambda)h} (x - \theta). \quad (110)$$

Parameterizing by $(\theta, \tau, \kappa, \lambda)$ isolates the invariant margin from the two dynamic decay rates. MOU-Peg fixes $\theta = 0$ and fixes τ at the training root-mean-square peg deviation. MOU-IFM fixes (θ, τ) at the training Gaussian margin estimates and maximizes over (κ, λ) . MOU-UQML maximizes the conditional criterion over all four parameters. The last fit remains a valid conditional transition kernel, but its long-run interpretation is accepted only if it passes the separate invariant-margin audit.

We also estimate a latent Gaussian state-space benchmark,

$$Z_{t+h} - \theta = e^{-\kappa h} (Z_t - \theta) + \eta_{t,h}, \quad X_t = Z_t + \varepsilon_t,$$

by the exact scalar Kalman likelihood. The innovations have the OU variance $\tau^2(1 - e^{-2\kappa h})$, and the observation errors are independent Gaussian noise. This benchmark asks whether apparent rapid correction is primarily a high-frequency measurement effect.

5.3 Parameter estimates and identification

MOU-IFM estimates $\hat{\kappa} = 0.884$ and $\hat{\lambda} = 2.353$. Its base conditional-mean half-life is about 1,130 minutes, whereas the combined half-life from $\kappa + \lambda$ is about 308 minutes. MOU-Peg gives nearly

Table 1: Central model estimates, baseline sample

Model	θ	τ	κ	λ	Hessian cond.	Conv.
OUF	0.000128	0.000826	1.361	—	3008741.5	True
MOU-Peg	0.000000	0.000739	0.878	2.314	66.5	True
MOU-IFM	0.000067	0.000736	0.884	2.353	77.5	True
MOU-UQML	0.000766	0.004241	0.026	1.861	75123.5	True

Notes: Rates are per day. OUF has no reset parameter. Fits use 120,000 systematically spaced exact five-minute transitions. The condition number is reported in the optimizer coordinates; all seven MOU starts reached the same objective to the pre-specified tolerance.

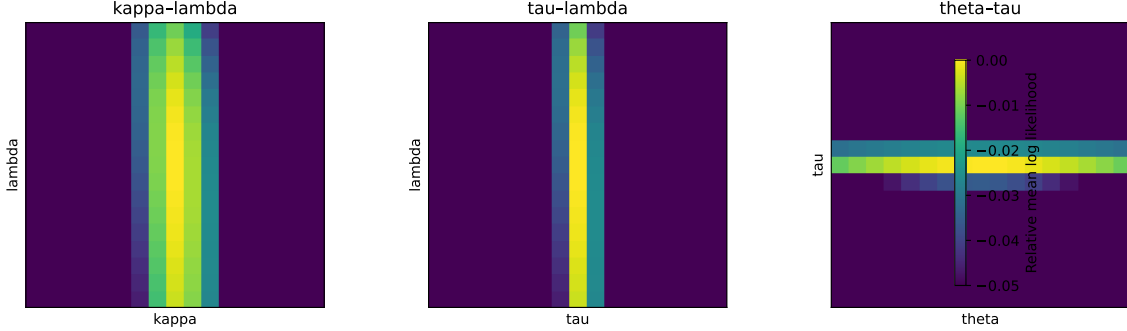


Figure 1: MOU-UQML identification slices. Colors are relative mean log-likelihoods; nuisance coordinates are fixed at the unrestricted estimate.

identical dynamic estimates. The unrestricted criterion instead moves toward a slow base OU, $\hat{\kappa} = 0.026$, a wide invariant margin, and $\hat{\lambda} = 1.861$. The repeated solutions and finite Hessian condition number show that this result is not merely a failed numerical optimization; it is the conditional criterion’s preferred contamination representation.

Figure 1 reports likelihood slices in the (κ, λ) , (τ, λ) , and (θ, τ) planes with the remaining coordinates held at the MOU-UQML estimate. These are diagnostic slices, not nuisance-reoptimized profile likelihoods, and are interpreted accordingly.

5.4 Invariant-margin audit

Table 2 and Figure 2 make the central distinction visible. MOU-UQML has an invariant standard deviation of 42.41 basis points, 5.76 times the empirical 7.36 basis points. It assigns 24.60 percent probability outside ± 50 basis points, compared with 0.157 percent in the data. It therefore fails structural interpretation by a wide margin. MOU-Peg and MOU-IFM match the empirical scale, and the latent noise OU implies an observed scale of 7.50 basis points, but all Gaussian margins fail the tail component of the audit. The repair required by the evidence is consequently twofold: constrain or separately estimate the margin, and allow that margin to be thick-tailed or semiparametric.

The state-space benchmark estimates a latent invariant standard deviation of 7.34 basis points and an observation-error standard deviation of 1.56 basis points. Observation noise accounts for 4.34 percent of the stationary variance. It is economically and statistically nonzero in scale, but is too small to reconcile a Gaussian stationary law with the empirical tails.

5.5 Out-of-sample conditional scores

Both constrained mixed models improve the full conditional log score relative to OUF in 2024 and 2025. MOU-IFM is marginally stronger than MOU-Peg, so the data do not require imposing

Table 2: Invariant-margin audit

Model	Mean (bp)	SD (bp)	KS	$\Pr(X > 50\text{bp})$ (%)	Gate
Data	0.671	7.359	0.000	0.157	–
OUF	1.280	8.255	0.189	0.000	Fail
MOU-Peg	0.000	7.390	0.171	0.000	Fail
MOU-IFM	0.671	7.359	0.141	0.000	Fail
MOU-UQML	7.657	42.405	0.439	24.598	Fail
LatentNoiseOU-observed	0.656	7.503	0.144	0.000	Fail

Notes: Model rows use the Gaussian invariant margin implied by the baseline estimate. The gate jointly checks mean displacement, scale, KS distance, and the 50-basis-point tail when at least 20 empirical exceedances are available. A failed gate blocks stationary structural interpretation; it does not imply that the conditional transition density is mathematically invalid.

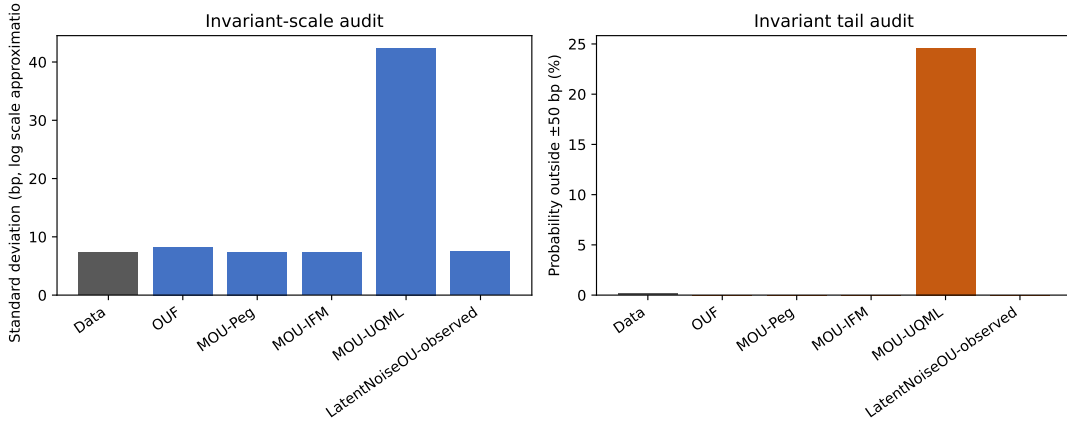


Figure 2: Invariant scale and 50-basis-point tail probabilities.

an exact zero log peg to obtain the dynamic gain. MOU-UQML scores still better, but the incremental gain is inseparable from the failed invariant margin. We therefore treat MOU-IFM as the primary interpretable Gaussian-margin estimate, MOU-Peg as a peg robustness check, and MOU-UQML as a conditional contamination diagnostic.

5.6 Boundary bootstrap and scope of inference

The null $\lambda = 0$ lies on the boundary. We simulate complete paths from the refitted OU null over 101 selected continuous five-minute segments, preserving every selected segment length. Each replication refits OUF, MOU-IFM, and MOU-UQML with the same estimators. We consider both stationary OU initial values and the empirical first observation of each segment; failed or nonconverged fits are retained and counted conservatively.

The observed LR is 8,081.46, whereas the simulated 95th percentiles are numerically close to zero under both initializations. The minimum attainable corrected pilot p-value is 0.01. This is a strong signal that the result is not created by conditioning on observed starting states, but it is not yet a formal boundary-test result because the replication gate is unmet.

5.7 Theory–evidence consistency

The empirical findings are consistent with the exact theory in a qualified sense. Equation (110) implements the additional conditional-mean decay proved in Section 4, and the constrained estimates find a positive, reproducible reset-like channel. The theory also says that a stationary reset mixture preserves its chosen invariant margin. The MOU-UQML failure is therefore not

Table 3: Five-minute out-of-sample scores

Evaluation period	Model	Mean full log score	Mean model-Copula score
validation 2024	OUF	8.21352	2.29856
validation 2024	MOU-Peg	8.42593	2.54116
validation 2024	MOU-IFM	8.42630	2.55920
validation 2024	MOU-UQML	8.43646	4.12606
test 2025	OUF	8.28503	2.12416
test 2025	MOU-Peg	8.52308	2.49281
test 2025	MOU-IFM	8.52348	2.50660
test 2025	MOU-UQML	8.54041	4.22616

Notes: All models are estimated on 2021–2023. The model-Copula score subtracts the model’s own invariant marginal log density from its transition log density. Because each model induces different probability integral transforms, it must be read jointly with Table 2, not as a stand-alone structural ranking.

Table 4: Continuous-path OU-null bootstrap pilot

Initialization	B	Observed LR	Null 95%	Corrected p	Formal gate
empirical segment initial	99	8081.46	0.0000	0.010	No
stationary initial	99	8081.46	0.0000	0.010	No

Notes: 25,000 transitions are used on the observed and simulated sides. The corrected p -value includes the finite-replication correction and would count a failed or nonconverged replication as an exceedance. No failures occurred. The pre-specified formal requirement is $B \geq 1999$ for each initialization; the present $B = 99$ experiment is a computational pilot.

a contradiction of that theorem: it shows that unrestricted conditional estimation selected the wrong invariant margin for structural use. Finally, the Gaussian margin failures do not invalidate the Markov–Copula construction. They identify the next model extension: retain the estimated dependence mechanism while replacing the Gaussian margin by a thick-tailed parametric or training-only semiparametric margin.

6 Conclusion

This paper develops a Copula-based theory of mean-reverting Markov processes and evaluates it on Kraken USDT/USD. The theory distinguishes a local drift restriction, an exact finite-horizon conditional-mean identity, and moment convergence under an initial law. It also separates a Copula CDF identity, which is pointwise, from density and conditional-kernel identities, which are naturally almost-everywhere statements. Monotone diffusion transforms yield observable densities only when invertibility, support, and Jacobian conditions hold. Conversely, the transform PDE follows from the semimartingale argument first along the law of the state process and becomes a pointwise PDE only under additional continuity and support conditions.

The stationary-reset family is the principal construction. A multiplicative mixture of a stationary transition kernel and its invariant projection is a consistent Markov evolution. It preserves the invariant margin and multiplies the base conditional-mean persistence by the no-reset probability. Under negative base drift and nonnegative reset intensity, this produces a strictly smaller absolute conditional-mean deviation whenever the integrated reset intensity is positive. The construction is related to stochastic resetting, but its contribution here is the Markov–Copula representation and the exact link among margins, observable densities, and mean reversion.

The empirical evidence supports the dependence mechanism more strongly than it supports any current stationary margin. MOU-IFM and MOU-Peg improve five-minute validation and

test density scores relative to OU and produce stable estimates of an additional decay channel. MOU-UQML scores still better, but does so with an invariant scale 5.76 times the empirical scale and an implausible 50-basis-point tail probability. It is therefore retained as a conditional contamination diagnostic, not a structural model. The margin-constrained Gaussian fits repair its mean and scale failure, while the invariant audit shows that they still miss the extreme kurtosis. A latent OU measurement-noise model attributes only 4.3 percent of stationary variance to observation noise and does not remove that discrepancy.

The appropriate conclusion is consequently limited. The data provide directional, finite-horizon evidence for an additional reset-like dependence channel. They do not yet validate global strong reversion, a Gaussian stationary law, or a universal forecasting advantage. Formal boundary inference also awaits the pre-specified 1,999-replication continuous-path bootstrap. The next empirical step is to combine the same coherent mixed Copula transition mechanism with thick-tailed or training-only semiparametric margins, then repeat the invariant audit, path bootstrap, price-proxy and frequency checks, and external-market validation.

A Detailed proofs

This appendix expands the measure-theoretic steps that are easy to hide inside phrases such as “interchange the limit and the conditional expectation.” It covers the general results used repeatedly in the main text. The model-specific OU, CIR, and special-function calculations remain next to their statements because they are direct substitutions into the displayed differential equations.

A.1 Copula kernels, density versions, and partial derivatives

Detailed proof of the kernel and derivative claims in Proposition 2.2. Fix $s < t$ and suppress the time subscripts temporarily. For the selected jointly Borel density c , define

$$K(u, B) = \int_B c(u, v) dv, \quad H^c(u, y) = K(u, (0, y]) = \int_0^y c(u, v) dv.$$

Row normalization makes $B \mapsto K(u, B)$ a probability measure for every u . Joint measurability of c , first for intervals and then by the monotone class theorem for all $B \in \mathcal{B}(I)$, makes $u \mapsto K(u, B)$ Borel. Column normalization and Tonelli’s theorem give, for every Borel B ,

$$\int_I K(u, B) du = \int_B \left\{ \int_I c(u, v) du \right\} dv = \text{Leb}(B), \quad (111)$$

so the selected kernel preserves the uniform law.

For every $x, y \in [0, 1]$, the copula reconstructed from c satisfies

$$C(x, y) = \int_0^x H^c(a, y) da. \quad (112)$$

For fixed y , the function $H^c(\cdot, y)$ is measurable and bounded by one. The fundamental theorem for Lebesgue integrals therefore implies

$$\partial_1 C(u, y) = H^c(u, y) \quad \text{for Leb-a.e. } u. \quad (113)$$

This is an equality for each fixed y ; its exceptional set may initially depend on y . Because $H^c(u, \cdot)$ and every conditional-CDF version are right-continuous, equality at all rational y outside one null set extends to all $y \in [0, 1]$ outside that same source-state null set.

At any source state u where $a \mapsto H^c(a, y)$ is continuous, the pointwise upgrade follows directly from the integral difference quotient:

$$\begin{aligned} \frac{C(u+h, y) - C(u, y)}{h} - H^c(u, y) &= \frac{1}{h} \int_u^{u+h} \{H^c(a, y) - H^c(u, y)\} da, \\ \left| \frac{C(u+h, y) - C(u, y)}{h} - H^c(u, y) \right| &\leq \sup_{|a-u| \leq |h|} |H^c(a, y) - H^c(u, y)| \longrightarrow 0. \end{aligned} \quad (114)$$

Thus a continuity condition on the integrated density row, not merely existence of a Lebesgue²-a.e. density, is what upgrades (113) to a pointwise derivative identity. The modified partial Dini construction of Fang et al. (2020) supplies another pointwise CDF-valued version $\mathfrak{D}_1 C(u, \cdot)$ satisfying the same reconstruction formula. Applying (113) to both versions shows that $\mathfrak{D}_1 C(u, \cdot) = H^c(u, \cdot)$ for a.e. u , although equality can fail on a source-null set. These uses of absolute continuity and differentiation are instances of the results in Folland (1999, Chapter 3).

Now fix $0 \leq s < r < t$ and $u \in I$. Since (3) holds for a.e. v , equality of its integrals over an arbitrary $B \in \mathcal{B}(I)$ and Tonelli's theorem give

$$\begin{aligned} K_{s,t}(u, B) &= \int_B \int_I c_{s,r}(u, w) c_{r,t}(w, v) dw dv \\ &= \int_I c_{s,r}(u, w) \left\{ \int_B c_{r,t}(w, v) dv \right\} dw \\ &= \int_I K_{r,t}(w, B) K_{s,r}(u, dw). \end{aligned} \quad (115)$$

This proves the kernel Chapman–Kolmogorov equation for every selected source u and every Borel B . Conversely, if both sides are absolutely continuous in the target variable, equality for every B identifies their Radon–Nikodym densities only for a.e. v .

Finally, let $C = C_{s,r}$ and $D = C_{r,t}$. The functions

$$w \mapsto \int_0^x c_{s,r}(a, w) da, \quad w \mapsto \int_0^y c_{r,t}(w, b) db$$

are measurable a.e. representatives of $\partial_2 C(x, w)$ and $\partial_1 D(w, y)$, respectively. Substitution into (9), followed by Tonelli and density CK, gives

$$\begin{aligned} (C_{s,r} * C_{r,t})(x, y) &= \int_I \int_0^x c_{s,r}(a, w) da \int_0^y c_{r,t}(w, b) db dw \\ &= \int_0^x \int_0^y \left\{ \int_I c_{s,r}(a, w) c_{r,t}(w, b) dw \right\} db da \\ &= \int_0^x \int_0^y c_{s,t}(a, b) db da = C_{s,t}(x, y). \end{aligned} \quad (116)$$

All changes in integration order are justified by nonnegativity. Conversely, equality of these copula CDFs identifies their two-dimensional densities only Lebesgue²-a.e. and hence, by Fubini, yields conditional density composition only for a.e. source state. This proves why star-product consistency alone is weaker than the selected pointwise kernel condition in Assumption 2.1. $\square$

A.2 Short-time kernels acting on moving functions

Lemma A.1 (Moving-function kernel lemma). *Fix $t \geq 0$ and an interior state $u \in I$. Write*

$$K_h(u, dv) = c_{t,t+h}(u, v) dv, \quad h > 0,$$

so that $K_h(u, I) = 1$. The following statements hold.

(a) Suppose that

$$K_h(u, \{v : |v - u| \geq \delta\}) \longrightarrow 0 \quad \text{for every } \delta > 0. \quad (117)$$

If g is bounded and continuous at u , then

$$\int_I g(v) K_h(u, dv) \longrightarrow g(u). \quad (118)$$

(b) In addition to part (a), let $r_h : I \rightarrow \mathbb{R}$ be measurable and suppose g is bounded and continuous at u and

$$\|r_h - g\|_\infty \longrightarrow 0. \quad (119)$$

Then

$$\int_I r_h(v) K_h(u, dv) \longrightarrow g(u). \quad (120)$$

(c) For an unbounded g , let $w : I \rightarrow [1, \infty)$ be measurable and define

$$\|f\|_w = \sup_{v \in I} \frac{|f(v)|}{w(v)}.$$

If

$$\|r_h - g\|_w \longrightarrow 0, \quad (121)$$

$$\sup_{0 < h < h_0} \int_I w(v) K_h(u, dv) < \infty, \quad (122)$$

$$\int_I g(v) K_h(u, dv) \longrightarrow g(u), \quad (123)$$

then (120) still holds.

Proof. For part (a), split the integral at distance δ from u :

$$\begin{aligned} \left| \int_I g(v) K_h(u, dv) - g(u) \right| &\leq \int_I |g(v) - g(u)| K_h(u, dv) \\ &\leq \sup_{|v-u| < \delta} |g(v) - g(u)| + 2\|g\|_\infty K_h(u, \{|v - u| \geq \delta\}). \end{aligned}$$

First let $h \downarrow 0$ and use (117); then let $\delta \downarrow 0$ and use continuity of g at u . This proves (118), or equivalently $K_h(u, \cdot) \Rightarrow \delta_u$.

For part (b), row normalization gives

$$\begin{aligned} \left| \int_I r_h(v) K_h(u, dv) - g(u) \right| &\leq \int_I |r_h(v) - g(v)| K_h(u, dv) \\ &\quad + \left| \int_I g(v) K_h(u, dv) - g(u) \right| \\ &\leq \|r_h - g\|_\infty + \left| \int_I g(v) K_h(u, dv) - g(u) \right|. \end{aligned}$$

Both terms tend to zero by (119) and part (a).

For part (c), replace the first term in the preceding bound by

$$\int_I |r_h(v) - g(v)| K_h(u, dv) \leq \|r_h - g\|_w \int_I w(v) K_h(u, dv).$$

Equations (121)–(123) prove the claim. This weighted version is the relevant one for Normal, Student, or other quantiles that diverge at the endpoints of I . $\square$

Pointwise convergence $r_h(v) \rightarrow g(v)$ alone is not enough in this lemma: the measure $K_h(u, dv)$ also changes with h . Conditions (119) or (121)–(122) provide the uniform integrability needed for the moving integrand.

A.3 Full integral proof of the quantile-generator drift identity

Detailed proof of Proposition 2.22. Fix t and put $q_t(v) = q(t, v) = Q_t(v)$ and $\dot{q}_t(v) = \partial_t q(t, v)$. Assume that all displayed first moments are finite and that, for the state u under consideration,

$$\mathcal{A}_t q_t(u) = \lim_{h \downarrow 0} \frac{\int_I q_t(v) c_{t,t+h}(u, v) dv - q_t(u)}{h} \quad (124)$$

exists. Also assume either part (b) or part (c) of Lemma A.1 with

$$r_h(v) = \frac{q(t+h, v) - q(t, v)}{h}, \quad g(v) = \dot{q}_t(v). \quad (125)$$

These are explicit sufficient conditions for the limit-interchange clause in Proposition 2.22.

For a regular conditional kernel version, strict invertibility of q_t gives, for Leb-almost every u ,

$$P_{t,t+h}^X(q_t(u), B) = \int_I \mathbf{1}_B(q_{t+h}(v)) c_{t,t+h}(u, v) dv. \quad (126)$$

Consequently,

$$b_X(t, q_t(u)) = \lim_{h \downarrow 0} \frac{\int_I q_{t+h}(v) c_{t,t+h}(u, v) dv - q_t(u)}{h}. \quad (127)$$

Using $\int_I c_{t,t+h}(u, v) dv = 1$, the difference quotient in (127) is exactly

$$\int_I \frac{q_{t+h}(v) - q_t(v)}{h} c_{t,t+h}(u, v) dv \quad (128)$$

$$+ \frac{\int_I q_t(v) c_{t,t+h}(u, v) dv - q_t(u)}{h}. \quad (129)$$

Lemma A.1 and (125) show that (128) tends to $\dot{q}_t(u)$. Definition (124) shows that (129) tends to $\mathcal{A}_t q_t(u)$. Hence

$$b_X(t, Q_t(u)) = \partial_t q(t, u) + \mathcal{A}_t q(t, \cdot)(u) = (\partial_t + \mathcal{A}_t) q(t, u). \quad (130)$$

Because U_t is uniform, the statement is naturally a Leb-almost-everywhere statement in u . A continuous kernel version extends it to every interior state where the limits exist. Strict monotonicity of q_t then transfers the sign of the right-hand side to the physical state $x = q_t(u)$, which proves the sign assertion in Proposition 2.22 after the limiting-mean condition is imposed separately. $\square$

A.4 Conditional expectations and observable transition densities

Detailed proof of Theorems 2.7 and 2.21, and Proposition 2.8. For $s < t$, define

$$H_{s,t}(u) = \int_I q_t(v) c_{s,t}(u, v) dv.$$

Column normalization and Tonelli's theorem give

$$\begin{aligned} \int_I |H_{s,t}(u)| du &\leq \int_I \int_I |q_t(v)| c_{s,t}(u, v) dv du \\ &= \int_I |q_t(v)| \left\{ \int_I c_{s,t}(u, v) du \right\} dv \\ &= \int_I |q_t(v)| dv < \infty. \end{aligned}$$

For a bounded $\mathcal{F}_s$ -measurable random variable Z , the Markov property, first for bounded truncations of q_t and then by L^1 convergence, gives

$$\begin{aligned}\mathbb{E}[ZX_t] &= \mathbb{E}[Zq_t(U_t)] \\ &= \mathbb{E}\left[Z \int_I q_t(v) c_{s,t}(U_s, v) dv\right] = \mathbb{E}[ZH_{s,t}(U_s)].\end{aligned}$$

This is the defining test-function identity for

$$\mathbb{E}[X_t | \mathcal{F}_s] = H_{s,t}(U_s).$$

Since $X_s = q_s(U_s)$ and $U_s \sim \text{Leb}$, equality $H_{s,t}(U_s) = q_s(U_s)$ almost surely is equivalent to $H_{s,t}(u) = q_s(u)$ for Leb-almost every u . This proves both directions of Theorem 2.7 for the fixed pair (s, t) ; no common null set over the uncountable collection of time pairs is asserted. Replacing the last equality by

$$H_{s,t}(u) - m = \mathcal{R}_{s,t}\{q_s(u) - m\}$$

proves both directions of Theorem 2.21.

Under the density assumptions of Proposition 2.8, the substitution $v = F_t(y)$, $dv = p_t(y) dy$, in the conditional kernel gives

$$p_{s,t}^X(x, y) = c_{s,t}\{F_s(x), F_t(y)\}p_t(y).$$

The three required identities follow by explicit changes of variables:

$$\begin{aligned}\int_{E_t^\circ} p_{s,t}^X(x, y) dy &= \int_I c_{s,t}\{F_s(x), v\} dv = 1, \\ \int_{E_s^\circ} p_s(x) p_{s,t}^X(x, y) dx &= p_t(y) \int_I c_{s,t}\{u, F_t(y)\} du = p_t(y), \\ \int_{E_r^\circ} p_{s,r}^X(x, z) p_{r,t}^X(z, y) dz &= p_t(y) \int_I c_{s,r}\{F_s(x), w\} c_{r,t}\{w, F_t(y)\} dw \\ &= p_t(y) c_{s,t}\{F_s(x), F_t(y)\} = p_{s,t}^X(x, y).\end{aligned}$$

The first line holds for every selected x , the second for Lebesgue-a.e. y , and the third for every selected x and Lebesgue-a.e. y . The final equality is precisely the density Chapman–Kolmogorov identity; integrating it over an arbitrary target Borel set yields the corresponding pointwise kernel identity. $\square$

A.5 The generator martingale theorem and rescaling PDE

Detailed proof of Theorem 2.10. Write

$$B(t, u) = (\partial_t + \mathcal{A}_t)q(t, u).$$

Membership in the time–space extended generator domain means that, after a localizing sequence if necessary,

$$L_t = q(t, U_t) - q(0, U_0) - \int_0^t B(s, U_s) ds \tag{131}$$

is a local martingale and the finite-variation integral is locally finite. If $B = 0$ for $dt \otimes \text{Leb}(du)$ -almost every (t, u) , uniformity of U_s and Tonelli’s theorem yield, for every $T < \infty$,

$$\mathbb{E} \int_0^T \mathbf{1}_{\{B(s, U_s) \neq 0\}} ds = \int_0^T \int_I \mathbf{1}_{\{B(s, u) \neq 0\}} du ds = 0.$$

Thus the integral in (131) vanishes and $q(t, U_t)$ is a local martingale. Class-(D) on each finite horizon makes the stopped family uniformly integrable, so the local martingale is a true martingale.

Conversely, suppose $q(t, U_t)$ is a true martingale. It is also a local martingale, and subtracting the local martingale L in (131) shows that

$$A_t := \int_0^t B(s, U_s) ds$$

is a local martingale. It is an absolutely continuous finite-variation process starting from zero. A finite-variation local martingale is indistinguishable from a constant, so $A_t = 0$ for all t almost surely and $B(s, U_s) = 0$ for $ds \otimes \mathbb{P}$ -almost every (s, ω) . Uniformity of U_s and Tonelli's theorem now give

$$0 = \int_0^T \mathbb{P}\{B(s, U_s) \neq 0\} ds = \int_0^T \int_I \mathbf{1}_{\{B(s, u) \neq 0\}} du ds.$$

Hence $B = 0$ for $dt \otimes \text{Leb}(du)$ -almost every (t, u) , proving the converse in Theorem 2.10. $\square$

Detailed generator proof of Proposition 2.20. For completeness, let

$$R_t = \mathcal{R}_{0,t} = \exp\left\{\int_0^t \gamma(a) da\right\}$$

and define

$$\tilde{q}(t, u) = \frac{q(t, u) - m}{R_t}.$$

Since R_t is deterministic, $R'_t = \gamma(t)R_t$, and $\mathcal{A}_t$ acts only on the state variable,

$$(\partial_t + \mathcal{A}_t)\tilde{q}(t, u) = \frac{1}{R_t} [(\partial_t + \mathcal{A}_t)q(t, u) - \gamma(t)\{q(t, u) - m\}]. \quad (132)$$

Thus the rescaled process is a martingale, subject to the preceding domain and class-(D) qualifications, exactly when (37) holds. At the finite-horizon level, multiplicativity gives $R_t = R_s \mathcal{R}_{s,t}$ and therefore

$$\mathbb{E}\left[\frac{X_t - m}{R_t} \middle| \mathcal{F}_s\right] = \frac{\mathcal{R}_{s,t}(X_s - m)}{R_s \mathcal{R}_{s,t}} = \frac{X_s - m}{R_s}.$$

The reverse calculation proves the converse part of Proposition 2.20 without appealing only to formal PDE algebra. $\square$

A.6 Diffusion transforms and reset mixtures

Detailed proof of Theorem 3.1. For Theorem 3.1, Itô's formula gives

$$\begin{aligned} d\{f(t, R_t) - m\} &= (\partial_t + \mathcal{L})f(t, R_t) dt + \sigma(R_t)f_x(t, R_t) dW_t \\ &= \gamma(t)\{f(t, R_t) - m\} dt + \sigma(R_t)f_x(t, R_t) dW_t. \end{aligned}$$

With $R_t^\gamma = \exp\{\int_0^t \gamma(a) da\}$, the deterministic product rule yields

$$d\left[\frac{f(t, R_t) - m}{R_t^\gamma}\right] = \frac{\sigma(R_t)f_x(t, R_t)}{R_t^\gamma} dW_t. \quad (133)$$

Condition (53) makes the stochastic integral in (133) square integrable on every finite horizon. Taking the conditional expectation between s and t and using $R_t^\gamma/R_s^\gamma = \mathcal{R}_{s,t}$ gives (54). Strict monotonicity gives the time-dependent inverse $R_s = f^{-1}(s, Y_s)$ and the transition kernel

$$P_{s,t}^Y(y, B) = P_{s,t}^R(f^{-1}(s, y), f^{-1}(t, B)),$$

which proves the Markov claim. Conversely, the conditional-mean identity makes the left side of (133) a martingale; write the PDE residual as

$$H(t, x) = (\partial_t + \mathcal{L})f(t, x) - \gamma(t)\{f(t, x) - m\}.$$

Subtracting the local-martingale part in Itô's formula shows that $A_t = \int_0^t H(s, R_s) ds$ is both a finite-variation process and a local martingale. Hence A is indistinguishable from zero and $H(t, R_t) = 0$ for $dt \otimes \mathbb{P}$ -almost every (t, ω) . Tonelli's theorem gives, for each $T < \infty$,

$$0 = \int_0^T \mathbb{P}\{H(t, R_t) \neq 0\} dt = \int_0^T \int_E \mathbf{1}_{\{H(t, x) \neq 0\}} \text{Law}(R_t)(dx) dt,$$

which is precisely (55). If H is continuous and $\text{Law}(R_t)$ has full support, $H(t, \cdot) = 0$ for almost every t ; time continuity then yields the pointwise PDE at all remaining times. Without these support and continuity conditions, semimartingale uniqueness alone does not justify a pointwise PDE on states that the diffusion does not visit. $\square$

Detailed proof of Theorems 4.1 and 4.2. Finally, write $a = a_{s,r}$ and $b = a_{r,t}$. Invariance gives $P_{s,r}\Pi = \Pi P_{r,t} = \Pi$ and $\Pi^2 = \Pi$, so the reset-kernel composition in Theorem 4.1 expands as

$$\begin{aligned} & \{aP_{s,r} + (1-a)\Pi\}\{bP_{r,t} + (1-b)\Pi\} \\ &= abP_{s,t} + \{a(1-b) + (1-a)b + (1-a)(1-b)\}\Pi \\ &= abP_{s,t} + (1-ab)\Pi = P_{s,t}^{\text{mix}}, \end{aligned}$$

because $ab = a_{s,t}$. For the Copula densities the corresponding integral is

$$\begin{aligned} & \int_I \{ac_{s,r}(u, w) + 1 - a\}\{bc_{r,t}(w, v) + 1 - b\} dw \\ &= abc_{s,t}(u, v) + a(1-b) + (1-a)b + (1-a)(1-b) \\ &= a_{s,t}c_{s,t}(u, v) + 1 - a_{s,t}, \end{aligned}$$

where row normalization, column normalization, and the base Chapman–Kolmogorov identity were used term by term. This density equality is for every selected u and a.e. v ; its integrated kernel equality is for every u and every Borel target set. Moreover, $a_{t,t+h} = 1 - \lambda(t)h + o(h)$ and $P_{t,t+h}h_0 = h_0 + h\mathcal{G}_th_0 + o(h)$ on the generator domain. Therefore

$$\frac{P_{t,t+h}^{\text{mix}}h_0 - h_0}{h} = \mathcal{G}_th_0 + \lambda(t)\{\Pi h_0 - h_0\} + o(1),$$

which proves (104). The conditional centered first moment is, directly in integral form,

$$\begin{aligned} \int_E (y - m)P_{s,t}^{\text{mix}}(x, dy) &= a_{s,t} \int_E (y - m)P_{s,t}(x, dy) \\ &\quad + (1 - a_{s,t}) \int_E (y - m)\pi(dy) \\ &= a_{s,t}\mathcal{R}_{s,t}(x - m), \end{aligned}$$

and its right derivative at $t = s$ gives $b_{\text{mix}}(s, x) = b_0(s, x) + \lambda(s)(m - x)$. $\square$

B Empirical Reproducibility and Deferred Gates

The raw file is checked against the SHA-256 digest recorded in the replication configuration. All timestamps are UTC; missing bars are not imputed; and exact-horizon pair construction never bridges a gap. The full commands are:

```
python scripts/17_major_revision_cycle.py --force
python scripts/18_major_revision_path_bootstrap.py --force
python scripts/19_build_submission_paper.py
```

The active tables are generated from CSV files rather than manually copied numbers. The current formal blockers are: (i) path-bootstrap replications below 1,999 per initialization, (ii) failure of every Gaussian invariant margin at the thick-tail gate, (iii) diagnostic likelihood slices rather than fully nuisance-reoptimized profiles, and (iv) incomplete price-proxy, frequency, and external-market checks. These blockers restrict the wording of the abstract and conclusion; they do not alter the mathematical validity of the transition-kernel construction.

References

- Rashad Ahmed, Iñaki Aldasoro, and Chanelle Duley. Public information and stablecoin runs. BIS Working Papers 1164, Bank for International Settlements, 2025. URL <https://www.bis.org/publ/work1164.htm>.
- Yacine Aït-Sahalia. Testing continuous-time models of the spot interest rate. *The Review of Financial Studies*, 9(2):385–426, 1996. doi: 10.1093/rfs/9.2.385.
- Yacine Aït-Sahalia. Maximum likelihood estimation of discretely sampled diffusions: A closed-form approximation approach. *Econometrica*, 70(1):223–262, 2002. doi: 10.1111/1468-0262.00274.
- Lennart Ante, Ingo Fiedler, Jan Willruth, and Fred Steinmetz. A systematic literature review of empirical research on stablecoins. *FinTech*, 2(1):3–47, 2023. doi: 10.3390/fintech2010002.
- Philippe Bergault, Louis Bertucci, David Bouba, Olivier Guéant, and Julien Guilbert. Automated market making: The case of pegged assets, 2024.
- Enrico Bibbona, Laura Sacerdote, and Emiliano Torre. A copula-based method to build diffusion models with prescribed marginal and serial dependence. *Methodology and Computing in Applied Probability*, 18(3):765–783, 2016. doi: 10.1007/s11009-016-9487-6.
- Bo Martin Bibby and Michael Sørensen. Martingale estimation functions for discretely observed diffusion processes. *Bernoulli*, 1(1/2):17–39, 1995. doi: 10.2307/3318679.
- K. C. Chan, G. Andrew Karolyi, Francis A. Longstaff, and Anthony B. Sanders. An empirical comparison of alternative models of the short-term interest rate. *The Journal of Finance*, 47(3):1209–1227, 1992. doi: 10.1111/j.1540-6261.1992.tb04011.x.
- Xiaohong Chen and Yanqin Fan. Estimation of copula-based semiparametric time series models. *Journal of Econometrics*, 130(2):307–335, 2006. doi: 10.1016/j.jeconom.2005.03.004.
- Xiaohong Chen, Wei Biao Wu, and Yanping Yi. Efficient estimation of copula-based semiparametric Markov models. *The Annals of Statistics*, 37(6B):4214–4253, 2009. doi: 10.1214/09-AOS719.
- John C. Cox, Jonathan E. Ingersoll, Jr., and Stephen A. Ross. A theory of the term structure of interest rates. *Econometrica*, 53(2):385–407, 1985. doi: 10.2307/1911242.
- William F. Darsow, Bao Nguyen, and Elwood T. Olsen. Copulas and Markov processes. *Illinois Journal of Mathematics*, 36(4):600–642, 1992. doi: 10.1215/ijm/1255987328.
- Arnaud Djogbenou, Efe Inan, and Joann Jasiak. Time-varying coefficient DAR model and stability measures for stablecoin prices: An application to Tether. *Journal of International Money and Finance*, 139:102946, 2023. doi: 10.1016/j.jimonfin.2023.102946.
- Kun Duan and Andrew Urquhart. The instability of stablecoins. *Finance Research Letters*, 52: 103573, 2023. doi: 10.1016/j.frl.2022.103573.
- Barry Eichengreen, My T. Nguyen, and Ganesh Viswanath-Natraj. Stablecoin devaluation risk. *The European Journal of Finance*, 31(11):1469–1496, 2025. doi: 10.1080/1351847X.2025.2505757.
- Stewart N. Ethier and Thomas G. Kurtz. *Markov Processes: Characterization and Convergence*. Wiley Series in Probability and Statistics. John Wiley & Sons, New York, 1986. doi: 10.1002/9780470316658.

- Martin R. Evans and Satya N. Majumdar. Diffusion with stochastic resetting. *Physical Review Letters*, 106(16):160601, 2011. doi: 10.1103/PhysRevLett.106.160601.
- Martin R. Evans, Satya N. Majumdar, and Grégoire Schehr. Stochastic resetting and applications. *Journal of Physics A: Mathematical and Theoretical*, 53(19):193001, 2020. doi: 10.1088/1751-8121/ab7cfe.
- Jun Fang, Fan Jiang, Yong Liu, and Jingping Yang. Copula-based Markov process. *Insurance: Mathematics and Economics*, 91:166–187, 2020. doi: 10.1016/j.insmatheco.2020.01.006.
- Gerald B. Folland. *Real Analysis: Modern Techniques and Their Applications*. John Wiley & Sons, New York, 2 edition, 1999. ISBN 978-0-471-31716-6.
- Vladimír Holý and Petra Tomanová. Estimation of Ornstein–Uhlenbeck process using ultra-high-frequency data with application to intraday pairs trading strategy, 2018.
- Cho-Hoi Hui, Andrew Wong, and Chi-Fai Lo. Stablecoin price dynamics under a peg-stabilising mechanism. *Journal of International Money and Finance*, 152:103280, 2025. doi: 10.1016/j.jimonfin.2025.103280.
- Olav Kallenberg. *Foundations of Modern Probability*, volume 99 of *Probability Theory and Stochastic Modelling*. Springer, Cham, 3 edition, 2021. doi: 10.1007/978-3-030-61871-1.
- Ioannis Karatzas and Steven E. Shreve. *Brownian Motion and Stochastic Calculus*, volume 113 of *Graduate Texts in Mathematics*. Springer, New York, 2 edition, 1991. doi: 10.1007/978-1-4612-0949-2.
- Mathieu Kessler. Estimation of an ergodic diffusion from discrete observations. *Scandinavian Journal of Statistics*, 24(2):211–229, 1997. doi: 10.1111/1467-9469.00059.
- Jiageng Liu, Igor Makarov, and Antoinette Schoar. Anatomy of a run: The Terra Luna crash. Working Paper 31160, National Bureau of Economic Research, 2023.
- Richard K. Lyons and Ganesh Viswanath-Natraj. What keeps stablecoins stable? *Journal of International Money and Finance*, 131:102777, 2023. doi: 10.1016/j.jimonfin.2022.102777.
- Yiming Ma, Yao Zeng, and Anthony Lee Zhang. Stablecoin runs and the centralization of arbitrage. Working paper, 2023.
- Roger B. Nelsen. *An Introduction to Copulas*. Springer Series in Statistics. Springer, New York, 2 edition, 2006. doi: 10.1007/0-387-28678-0.
- Frank W. J. Olver, Daniel W. Lozier, Ronald F. Boisvert, and Charles W. Clark, editors. *NIST Handbook of Mathematical Functions*. Cambridge University Press, Cambridge, 2010. ISBN 978-0-521-19225-5.
- Philip E. Protter. *Stochastic Integration and Differential Equations*, volume 21 of *Stochastic Modelling and Applied Probability*. Springer, Berlin, 2 edition, 2005. doi: 10.1007/978-3-662-10061-5.
- Oldrich Vasicek. An equilibrium characterization of the term structure. *Journal of Financial Economics*, 5(2):177–188, 1977. doi: 10.1016/0304-405X(77)90016-2.